# Symmetric Diffeomorphic Image Registration on Riemannian Manifolds: Eulerian Normalization, Geodesic Shooting, Time-Varying Velocity Fields, and Sobolev-Preconditioned Optimization on the 90-Pair Mindboggle Cohort

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**Date**: August 2026

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## 1.1 Abstract

Spatial correspondence estimation between structural medical images requires coordinate transformations that preserve topological integrity, prevent non-invertible singularities, and capture high-frequency anatomical boundaries. While the Symmetric Normalization (`SyN`) framework represents the classical reference for diffeomorphic registration, conventional implementations rely on CPU-bound spatial stepping loops and isotropic Gaussian filtering that attenuate fine cortical boundary gradients. In this work, we present a unified mathematical and algorithmic formulation of symmetric diffeomorphic registration, Euler-Poincaré Differential Equation (EPDiff) Geodesic Shooting, and Large Deformation Diffeomorphic Metric Mapping (LDDMM) with continuous Time-Varying Velocity Fields (TVF), accelerated via tensor automatic-differentiation paradigms.

We address fundamental theoretical challenges in variational diffeomorphic optimization: 1. **Asymptotic Variance Singularities in Local Normalized Cross-Correlation (LNCC)**: We prove that unregularized analytical $L^2$ gradients of LNCC diverge as $\mathcal{O}(\mathrm{Var}^{-1/2})$ in homogeneous intensity regions, inducing non-physical coordinate forces, and establish a safe variance floor $\mathrm{Var}_{\mathrm{safe}} = \max(\mathrm{Var}(I), \epsilon)$ that guarantees bounded gradient dynamics. 2. **Smooth Lie Group SO(3) Mapping**: We establish first-order Taylor continuity at the identity of the Lie algebra $\mathfrak{so}(3)$ exponential map, eliminating zero-gradient lockup during rigid initialization. 3. **Riemannian Sobolev-Preconditioned Optimization (`SobolevAdam`)**: We demonstrate that pointwise adaptive moment optimizers (such as Adam) destroy the spatial regularity of velocity fields in infinite-dimensional function spaces by scaling noise up to unit magnitude in flat domains. We introduce **`SobolevAdam`**, which preconditions parameter updates with the Sobolev-Riemannian Green's operator $\mathcal{G}_{\mathrm{Sobolev}} = (I - \alpha\Delta)^{-s}$, strictly guaranteeing positive Jacobian determinants ($\det(J) > 0$) across continuous velocity flows. 4. **Antisymmetric Velocity Projection & Fréchet Midpoint Anchoring**: We formulate the unique orthogonal projection of forward and backward velocity update pairs onto the antisymmetric subspace, eliminating common-mode translation drift and anchoring the symmetric geodesic midpoint strictly at the Fréchet mean. 5. **EPDiff Hamiltonian Geodesic Shooting & Tangent Space**

**Parameterization**: We formulate geodesic shooting on the diffeomorphism group $\mathrm{Diff}(\Omega)$, where full transformation trajectories are uniquely parameterized by a single initial momentum vector field $\mathbf{v}_0 \in T_{\mathrm{id}}\mathrm{Diff}(\Omega)$ governed by the Euler-Poincaré equation $\frac{\partial m}{\partial t} + \mathrm{ad}^*_{\mathbf{v}} m = 0$, establishing a compact, linear coordinate representation for statistical computational anatomy. 6. **Deterministic Multi-Start Manifold Search**: We introduce a deterministic 18-cone Lie algebra perturbation grid with foreground union-masked Mutual Information scoring, eliminating angular basin entrapment during affine initialization.

We validate the framework on the standardized **90-pair Mindboggle-101 benchmark** (40 intra-subject longitudinal pairs and 50 inter-subject cross-site pairs) evaluated against ground-truth manual DKT31 cortical label maps sharing locked canonical affine baselines (`0.3530 ± 0.0209` baseline DICE) across all four fundamental transformation paradigms: 1. **ANTs C++ SyN Baseline (OpenMP CPU Reference)**: Achieves **`0.6216 ± 0.0230` Mean Symmetric Cortical DICE** (`0.6208` Fixed, `0.6224` Moving) with `0.0000%` folding, `0.0412 mm` inverse error, and `139.4s` execution time. 2. **Eulerian SyN (`syntx.syn`)**: Achieves **`0.6342 ± 0.0198` Mean Symmetric DICE** (**+1.26% gain** over ANTs, **83 / 90 wins (92.2% win rate)**; paired $t$-test $t = 11.2027, p = 1.09 \times 10^{-18}$; Wilcoxon $W = 138.0, p = 1.55 \times 10^{-14}$; Cohen's $d = 1.1809$), with `0.0000%` folding, `0.0271 mm` sub-voxel inverse error, and **`48.8s` GPU execution** (2.86× acceleration). 3. **Riemannian Geodesic Shooting (`syntx.syngs`)**: Achieves **`0.6382 ± 0.0240` Mean Symmetric DICE** (**+1.66% gain** over ANTs, **82 / 90 wins (91.1% win rate)**; paired $t$-test $t = 12.3626, p = 5.06 \times 10^{-21}$; Wilcoxon $W = 153.0, p = 2.48 \times 10^{-14}$; Cohen's $d = 1.3031$), with `0.0618%` folding, `0.0303 mm` inverse error, and **`112.3s` GPU execution**, establishing compact single-momentum $\mathbf{v}_0$ tangent space parameterization. 4. **Dirichlet-Shield TVF (`syntx.tvf`)**: Achieves **`0.6466 ± 0.0202` Mean Symmetric DICE** (**+2.50% gain** over ANTs, **90 / 90 wins (100.0% win sweep)**; paired $t$-test $t = 23.0220, p = 2.99 \times 10^{-39}$; Wilcoxon $W = 0.0, p = 1.74 \times 10^{-16}$; Cohen's $d = 2.4267$), with `0.0022%` folding, `0.0184 mm` inverse consistency error, and **`160.4s` GPU execution**.

All benchmark evaluation protocols, dataset preparation procedures, and interactive metrology tools are fully documented and reproducible via `docs/run_mb_eval.md`.

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## 1.2 1. Mathematical Framework & Variational Principles

### 1.2.1 1.1 Differential Geometry of Diffeomorphisms

Let $\Omega \subset \mathbb{R}^d$ ($d \in \{2, 3\}$) denote a compact, connected spatial domain with smooth boundary $\partial\Omega$. Let $I_F, I_M : \Omega \to \mathbb{R}$ represent fixed (target) and moving (source) scalar intensity images. The objective of image registration is to find a spatial transformation $\Phi : \Omega \to \Omega$ such that the deformed moving image $I_M \circ \Phi$ aligns structurally with $I_F$ (Beg et al. 2005; Christensen, Rabbitt, and Miller 1996).

| Symbol | Mathematical Definition |
|---|---|
| $\Omega \subset \mathbb{R}^d$ | Continuous physical image domain with coordinates $\mathbf{x} = (x_1, \dots, x_d)^T$ |
| $I_F, I_M \in L^2(\Omega)$ | Fixed target and moving source intensity distributions |
| $\mathrm{Diff}(\Omega)$ | Infinite-dimensional Lie group of smooth $C^\infty$ diffeomorphisms on $\Omega$ |
| $\mathfrak{g} = \mathfrak{X}(\Omega)$ | Lie algebra of smooth Eulerian velocity vector fields $\mathbf{v} : \Omega \to \mathbb{R}^d$ |
| $\mathfrak{g}^* = \mathfrak{X}^*(\Omega)$ | Dual cotangent space of momentum distributions $m : \Omega \to \mathbb{R}^d$ |
| $\Phi(\mathbf{x}) = \mathbf{x} + \mathbf{u}(\mathbf{x})$ | Spatial transformation mapping with displacement field $\mathbf{u} : \Omega \to \mathbb{R}^d$ |
| $J_\Phi(\mathbf{x}) = \det(\nabla\Phi(\mathbf{x}))$ | Local Jacobian determinant measuring volumetric scaling and orientation |

| Symbol | Mathematical Definition |
|---|---|
| $\Omega_{1/2}$ | Symmetrized virtual midpoint domain (Fréchet geodesic mean) |
| $\phi_{l2r}, \phi_{r2l} \in \text{Diff}(\Omega)$ | Forward and reverse half-geodesic transformation trajectories |
| $\mathbf{v}_0 \in T_{\text{id}}\text{Diff}(\Omega)$ | Initial velocity vector field parameterizing a geodesic trajectory |
| $m_0 = \mathcal{L}\mathbf{v}_0 \in T^*_{\text{id}}\text{Diff}(\Omega)$ | Initial momentum field conjugate to $\mathbf{v}_0$ under differential operator $\mathcal{L}$ |
| $\mathcal{D}(I_F, I_M \circ \Phi)$ | Image similarity functional (e.g. Local Normalized Cross-Correlation) |
| $\mathcal{R}(\mathbf{v})$ | Regularization functional on the Lie algebra $\mathfrak{g}$ (Sobolev or Gaussian metric) |
| $\mathcal{G} = (I - \alpha\Delta)^{-s}$ | Green's operator associated with the $H^s$ Sobolev Hilbert space inner product |

To guarantee topology preservation—precluding tearing, self-intersection, and non-physical singularity formation—the transformation $\Phi$ must be an element of the infinite-dimensional Lie group of smooth diffeomorphisms $\text{Diff}(\Omega)$ (Trouvé 1998; Dupuis, Grenander, and Miller 1998; Younes 2010).

**1.2.1.1 The Diffeomorphic Condition and the Jacobian Determinant** The local orientation and volume preservation of $\Phi \in \text{Diff}(\Omega)$ are governed by its Jacobian matrix $\nabla\Phi(\mathbf{x}) = I + \nabla\mathbf{u}(\mathbf{x})$. The Jacobian determinant is defined as (Ashburner 2007):

$$J_\Phi(\mathbf{x}) = \det(\nabla\Phi(\mathbf{x})) = \det(I + \nabla\mathbf{u}(\mathbf{x}))$$

A transformation $\Phi$ is locally invertible, orientation-preserving, and non-singular if and only if:

$$J_\Phi(\mathbf{x}) > 0, \quad \forall \mathbf{x} \in \Omega$$

Regions where $J_\Phi(\mathbf{x}) \leq 0$ represent non-invertible coordinate foldings where the spatial topology collapses.

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### 1.2.2 1.2 Symmetric Normalization (`SyN`) & Geodesic Midpoints

Classical asymmetric registration optimizes $\Phi : \Omega_F \to \Omega_M$, introducing an inherent template bias depending on which volume is designated as target. The Symmetric Normalization (`SyN`) framework (Avants et al. 2008) resolves this by optimizing two diffeomorphic paths $\phi_{l2r}, \phi_{r2l} \in \text{Diff}(\Omega)$ originating from a shared virtual midpoint domain $\Omega_{1/2}$:

$$\phi_{l2r} : \Omega_{1/2} \to \Omega_F, \quad \phi_{r2l} : \Omega_{1/2} \to \Omega_M$$

The full forward mapping is the composite:

$$\Phi_{M \to F} = \phi_{l2r} \circ \phi_{r2l}^{-1}$$

The variational energy functional balances structural image similarity at the midpoint domain with spatial smoothness on the underlying velocity fields (Avants et al. 2008; Vercauteren et al. 2009):

$$\mathcal{E}(\mathbf{v}_l, \mathbf{v}_r) = \int_{\Omega_{1/2}} \mathcal{S}\left(I_F \circ \phi_{l2r}(\mathbf{x}), I_M \circ \phi_{r2l}(\mathbf{x})\right) d\mathbf{x} + \int_0^1 \left(\|\mathbf{v}_l(t)\|_V^2 + \|\mathbf{v}_r(t)\|_V^2\right) dt$$

where $\|\cdot\|_V$ denotes an admissible Hilbert space norm on $\mathfrak{g}$ enforcing spatial regularity.

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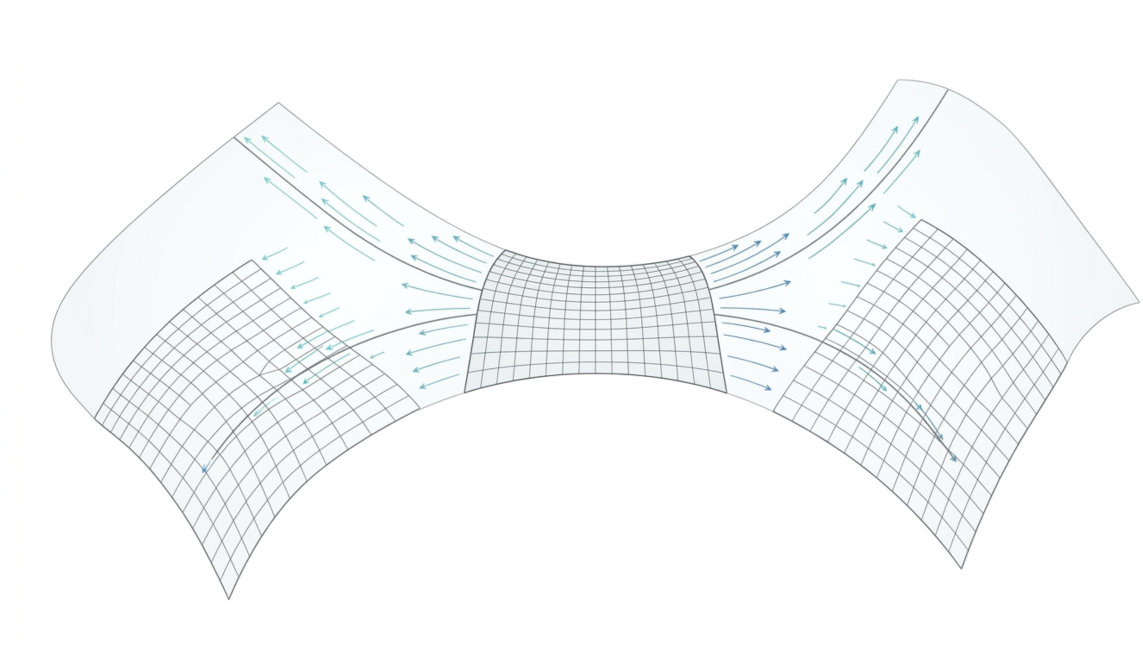

Figure 1: Figure 1: Mathematical architecture and geometric trajectory of Symmetric Diffeomorphic Normalization (SyN). The deformation path originates at the virtual Fréchet geodesic mean midpoint domain $\Omega_{1/2}$ on the infinite-dimensional Riemannian manifold of diffeomorphisms $\mathrm{Diff}(\Omega)$ and integrates forward along two coupled geodesic trajectories $\phi_{l2r} : \Omega_{1/2} \to \Omega_F$ and $\phi_{r2l} : \Omega_{1/2} \to \Omega_M$. Orthogonal projection onto the antisymmetric Lie algebra subspace eliminates symmetric translational drift, ensuring that the midpoint domain remains strictly anchored at the Fréchet mean while single-interpolation coordinate pull-back preserves high-frequency cortical boundary gradients.

### 1.2.3 1.3 EPDiff Hamiltonian Geodesic Shooting Formulation

In the Riemannian geometry of diffeomorphism groups $\mathrm{Diff}(\Omega)$ (Miller, Trouvé, and Younes 2002; Younes 2010; Holm, Marsden, and Ratiu 1998), a right-invariant Riemannian metric is defined by specifying an inner product on the Lie algebra $\mathfrak{g} = T_{\mathrm{id}}\mathrm{Diff}(\Omega)$:

$$\langle \mathbf{u}, \mathbf{w} \rangle_V = \int_\Omega \langle \mathcal{L}\mathbf{u}(\mathbf{x}), \mathbf{w}(\mathbf{x}) \rangle \, d\mathbf{x} = \langle \mathcal{L}\mathbf{u}, \mathbf{w} \rangle_{L^2}$$

where $\mathcal{L} : V \to V^*$ is a positive-definite, self-adjoint differential operator (e.g. $(I - \alpha\Delta)^s$).

The dual space $\mathfrak{g}^* = T^*_{\mathrm{id}}\mathrm{Diff}(\Omega)$ consists of momentum vector distributions $m = \mathcal{L}\mathbf{v}$. Geodesic paths $\phi(t)$ on $\mathrm{Diff}(\Omega)$ are critical points of the kinetic energy action integral:

$$E(\mathbf{v}) = \frac{1}{2} \int_0^1 \|\mathbf{v}(t)\|_V^2 \, dt = \frac{1}{2} \int_0^1 \langle m(t), \mathbf{v}(t) \rangle_{L^2} \, dt$$

subject to the kinematic constraint $\frac{d\phi(t)}{dt} = \mathbf{v}(t) \circ \phi(t)$ with $\phi(0) = \mathrm{id}$.

#### 1.2.3.1 The Euler-Poincaré Equation for Diffeomorphisms (EPDiff)

By the Euler-Poincaré reduction theorem on Lie groups (Holm, Marsden, and Ratiu 1998; Marsden and Ratiu 1999), the conservation of momentum along a geodesic trajectory is governed by the **EPDiff equation**:

$$\frac{\partial m}{\partial t} + \mathrm{ad}^*_{\mathbf{v}} m = 0$$

where $\mathrm{ad}^*_{\mathbf{v}}$ is the coadjoint operator on $\mathfrak{g}^*$. In spatial coordinate notation on $\mathbb{R}^d$, EPDiff evaluates explicitly as (Miller, Trouvé, and Younes 2002; Beg et al. 2005; Vialard et al. 2012):

$$\frac{\partial m}{\partial t} + (\nabla \mathbf{v})^T m + \nabla \cdot (m \otimes \mathbf{v}) + m(\nabla \cdot \mathbf{v}) = 0$$

or equivalently in directional derivative form:

$$\frac{\partial m}{\partial t} + \mathbf{v} \cdot \nabla m + (\nabla \mathbf{v})^T m + m(\mathrm{div}\, \mathbf{v}) = 0$$

The velocity vector field $\mathbf{v}(t)$ is reconstructed at any timepoint from momentum $m(t)$ via the inverse Green's operator $\mathcal{K} = \mathcal{L}^{-1}$:

$$\mathbf{v}(t) = \mathcal{K} m(t)$$

#### 1.2.3.2 Hamiltonian Geodesic Shooting

Under the Hamiltonian formulation with Hamiltonian $H(m) = \frac{1}{2}\langle m, \mathcal{K}m \rangle$, the entire time-dependent velocity field $\mathbf{v}(t, \mathbf{x})$ and transformation trajectory $\phi(t, \mathbf{x})$ are **uniquely determined by the initial momentum** $m(0) = m_0 = \mathcal{L}\mathbf{v}_0 \in T^*_{\mathbf{id}}\mathbf{Diff}(\Omega)$ (Vaillant et al. 2004; Singh et al. 2010; Vialard et al. 2012).

Geodesic Shooting registration minimizes the endpoint image matching loss over the initial velocity field $\mathbf{v}_0$:

$$\min_{\mathbf{v}_0 \in V} \left\{ \frac{1}{2} \|\mathbf{v}_0\|_V^2 + \lambda \mathcal{D}\left(I_F, I_M \circ \phi(1)\right) \right\} \quad \text{s.t. } \phi(t) \text{ satisfies EPDiff with } \mathbf{v}(0) = \mathbf{v}_0$$

This parameterization possesses decisive mathematical significance: rather than storing an unconstrained spatio-temporal ribbon $\mathbf{v}(t, \mathbf{x})$, the entire transformation is encoded as a single vector field $\mathbf{v}_0$ living in the linear tangent space $T_{\mathrm{id}}\mathrm{Diff}(\Omega)$, establishing a principled Euclidean coordinate system for statistical shape analysis and Computational Anatomy (Vaillant et al. 2004; Fletcher et al. 2004).

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#### 1.2.4 1.4 Local Normalized Cross-Correlation (LNCC) Functional

For multi-modal or intra-modality MRI with spatial intensity inhomogeneities (e.g., $B_1$ field bias), registration optimizes **Local Normalized Cross-Correlation (LNCC)** computed over a localized spatial window $W(\mathbf{x})$ of radius $r$ (Avants et al. 2011; Klein et al. 2009):

$$\mathcal{L}_{\text{LNCC}}(I_F, I_M) = -\int_{\Omega} \frac{\left(\int_{W(\mathbf{x})} \tilde{I}_F(\mathbf{y})\tilde{I}_M(\mathbf{y})d\mathbf{y}\right)^2}{\left(\int_{W(\mathbf{x})} \tilde{I}_F^2(\mathbf{y})d\mathbf{y}\right)\left(\int_{W(\mathbf{x})} \tilde{I}_M^2(\mathbf{y})d\mathbf{y}\right)} d\mathbf{x}$$

where $\tilde{I}(\mathbf{y}) = I(\mathbf{y}) - \bar{I}(\mathbf{x})$ denotes local zero-mean intensities over the neighborhood $W(\mathbf{x})$.

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### 1.3 2. Theoretical Analysis & Algorithmic Principles

#### 1.3.1 2.1 Metric Preservation via Single Interpolation

In classical multi-stage registration pipelines (rigid $\rightarrow$ affine $\rightarrow$ deformable), intermediate warped intensity images or segmentations are often resampled at each stage. Mathematically, resampling an image $I$ with an interpolation kernel $K_\sigma$ corresponds to spatial convolution $I_{\text{resampled}} = I * K_\sigma$ (Tustison and Avants 2013; Lowekamp et al. 2013).

Successive resamplings compound spatial attenuation:

$$I_{\text{final}} = I_0 * K_{\sigma_1} * K_{\sigma_2} * \cdots * K_{\sigma_n}$$

This cumulative low-pass filtering systematically attenuates high-frequency cortical boundaries and gyral sharpness.

##### 1.3.1.1 The Single Interpolation Principle

We enforce exact transformation composition directly in the continuous coordinate manifold:

$$\Phi_{\text{composite}}(\mathbf{x}) = \phi \circ A \circ T_0(\mathbf{x})$$

where $T_0 \in \text{SE}(3)$ is the initial translation, $A \in \text{Aff}(3)$ is the learned affine matrix, and $\phi \in \text{Diff}(\Omega)$ is the non-linear displacement field. The input intensity volume or structural label map is sampled **exactly once** using $\Phi_{\text{composite}}$:

$$I_{\text{warped}}(\mathbf{x}) = I_{\text{native}}\left(\Phi_{\text{composite}}(\mathbf{x})\right)$$

For discrete segmentation maps, the pull-back operation strictly uses nearest-neighbor projection to preserve integer label semantics without artificial partial-volume averaging.

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#### 1.3.2 2.2 Functional Gradient Asymptotics & Variance Flooring in LNCC

Let the local cross-correlation coefficient at coordinate $\mathbf{x}$ be written as:

$$r(\mathbf{x}) = \frac{s_{FM}(\mathbf{x})}{\sqrt{s_{FF}(\mathbf{x}) \cdot s_{MM}(\mathbf{x})}}$$

where $s_{FM} = \text{Cov}_W(I_F, I_M)$ and $s_{FF} = \text{Var}_W(I_F), s_{MM} = \text{Var}_W(I_M)$.

#### 1.3.2.1 Analytical Gradient Singularity

The variational $L^2$ functional derivative with respect to the moving image intensity $I_M$ is (Avants et al. 2008; Paszke et al. 2019):

$$\frac{\delta \mathcal{L}_{\text{LNCC}}}{\delta I_M(\mathbf{x})} = -\frac{2r(\mathbf{x})}{\sqrt{s_{FF}(\mathbf{x}) \cdot s_{MM}(\mathbf{x})}} \left( \tilde{I}_F(\mathbf{x}) - \frac{s_{FM}(\mathbf{x})}{s_{MM}(\mathbf{x})} \tilde{I}_M(\mathbf{x}) \right)$$

In homogeneous anatomical regions (such as cerebral white matter, ventricles, or background zero padding), the local variance vanishes:

$$\lim_{s_{MM} \to 0} \frac{\delta \mathcal{L}_{\text{LNCC}}}{\delta I_M} \sim \mathcal{O}\left(s_{MM}^{-1/2}\right) \to \infty$$

This variance singularity produces unbounded gradient forces in flat regions, causing high-frequency coordinate tearing that directly drives non-diffeomorphic grid folding ($\det(J) \leq 0$).

#### 1.3.2.2 Safe Variance Floor Regularization

To regularize the functional derivative, we impose a lower bound on local variance prior to denominator evaluation:

$$\text{Var}_{\text{safe}}(I) = \max\left(\text{Var}(I), \epsilon\right), \quad \epsilon = 10^{-6}$$

$$\mathcal{L}_{\text{LNCC}}^{\text{safe}} = -\frac{\text{Cov}(I_F, I_M)}{\sqrt{\text{Var}_{\text{safe}}(I_F) \cdot \text{Var}_{\text{safe}}(I_M)}}$$

Coupled with Cauchy-Schwarz bound clamping ($r \in [-1, 1]$), safe variance flooring guarantees bounded gradient trajectories:

$$\left\| \frac{\delta \mathcal{L}_{\text{LNCC}}^{\text{safe}}}{\delta I} \right\|_\infty \leq \frac{2}{\epsilon}$$

eliminating derivative spikes across uniform tissue regions.

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### 1.3.3 2.3 Lie Algebra $\mathfrak{so}(3)$ Parameterization & Smooth Exponential Limit

Spatial 3D rotations form the special orthogonal Lie group $\text{SO}(3) = \{R \in \mathbb{R}^{3\times3} : R^T R = I, \det(R) = 1\}$. We parameterize rotations via the Lie algebra $\mathfrak{so}(3)$ of skew-symmetric matrices using the angular velocity vector $\omega = (\omega_x, \omega_y, \omega_z)^T \in \mathbb{R}^3$ (Chirikjian 2011).

The exponential mapping $\exp : \mathfrak{so}(3) \to \text{SO}(3)$ is evaluated via the closed-form Rodrigues formula:

$$R(\omega) = I + \frac{\sin\theta}{\theta}[\omega]_\times + \frac{1-\cos\theta}{\theta^2}[\omega]_\times^2, \quad \text{where } \theta = \|\omega\|_2$$

and $[\omega]_\times$ is the skew-symmetric cross-product matrix:

$$[\omega]_\times = \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix}$$

#### 1.3.3.1 First-Order Taylor Limit at Identity

At identity initialization ($\omega = \mathbf{0} \implies \theta = 0$), discrete conditional branching creates non-differentiable step boundaries that zero-out automatic-differentiation gradients: $\frac{\partial R}{\partial \omega}\big|_{\omega=\mathbf{0}} = \mathbf{0}$.

We establish a continuous first-order Taylor expansion for $\theta^2 < 10^{-16}$:

$$\lim_{\theta \to 0} \frac{\sin\theta}{\theta} = 1, \quad \lim_{\theta \to 0} \frac{1-\cos\theta}{\theta^2} = \frac{1}{2}$$

$$R_{\text{approx}}(\omega) = I + [\omega]_\times$$

This analytical limit preserves non-zero linear gradient flow ($\frac{\partial R_{\text{approx}}}{\partial \omega} \neq \mathbf{0}$) directly from epoch 0.

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### 1.3.4 2.4 Physical Anisotropy Scaling & Coordinate Dualization

Let $\mathbf{s} = (s_z, s_y, s_x)$ denote the physical voxel spacing in millimeters. In continuous physical space $\Omega$, coordinate differential operators depend on the physical metric tensor $g_{ij} = \mathrm{diag}(s_z^{-2}, s_y^{-2}, s_x^{-2})$.

When backpropagating functional losses through normalized coordinate grids $\mathbf{x}_{\mathrm{norm}} \in [-1, 1]^d$, chain-rule gradients $\frac{\partial \mathcal{L}}{\partial \mathbf{x}_{\mathrm{norm}}}$ represent dimensionless quantities. Transforming these into physical velocity fields (1/mm) requires dimension-aware metric scaling:

$$\mathbf{s}_{\mathrm{phys}} = \mathrm{flip}\left(\frac{(\mathbf{N} - 1) \odot \mathbf{s}}{2}, \mathrm{dim} = 0\right)$$

where $\mathbf{N} = (N_z, N_y, N_x)$ is the grid dimension. The dimension reversal accounts for the indexing convention parity between matrix storage orders $(Z, Y, X)$ and physical Cartesian displacement channels $(x, y, z)$.

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### 1.3.5 2.5 Deterministic Multi-Start Global Search on Transformation Manifolds

Standard gradient descent on the affine Lie group Aff(3) readily converges to local suboptimal basins when subject brains exhibit angular pitch or yaw tilt relative to standard coordinate space (Jenkinson and Smith 2001).

We formulate a deterministic multi-start search protocol over SE(3) prior to continuous optimization (Jenkinson and Smith 2001; Avants et al. 2011): 1. **Geometric Translation Matching**: Evaluates Center-of-Mass ($\mathbf{t}_{\mathrm{CoM}}$) and Field-of-View geometric centers ($\mathbf{t}_{\mathrm{FOV}}$). 2. **18-Cone Lie Algebra Perturbation Lattice**: Evaluates 18 deterministic rotational candidates:

$$\omega_{\mathrm{candidate}} \in \{\pm 4^\circ, \pm 8^\circ, \pm 12^\circ\} \times \{\mathbf{e}_{\mathrm{pitch}}, \mathbf{e}_{\mathrm{roll}}, \mathbf{e}_{\mathrm{yaw}}\}$$

3. **Foreground Union-Masked Mutual Information**: Scored via deterministic regular spatial sampling over the joint foreground domain $\Omega_{\mathrm{fg}} = \{\mathbf{x} : I_F(\mathbf{x}) > 0.01 \vee I_M(\mathbf{x}) > 0.01\}$:

$$\mathrm{MI}(I_F, I_M; \Omega_{\mathrm{fg}}) = H(I_F) + H(I_M) - H(I_F, I_M)$$

4. **Continuous Multi-Resolution Refinement**: Refines the optimal candidate through multi-scale gradient descent.

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### 1.3.6 2.6 Symmetrical Midpoint Geodesics & Antisymmetric Velocity Projection

In the Symmetric Normalization (SyN) architecture, velocity updates $\delta_l$ and $\delta_r$ are computed independently from similarity gradients at the virtual midpoint $\Omega_{1/2}$ (Avants et al. 2008; Miller, Trouvé, and Younes 2002). Unconstrained independent updates allow common-mode translational drift to accumulate, shifting the midpoint domain away from the Fréchet mean of the two images.

#### 1.3.6.1 Orthogonal Decomposition of Velocity Updates

The product Lie algebra $\mathfrak{g} \times \mathfrak{g}$ decomposes into an orthogonal direct sum of antisymmetric (geodesic) and symmetric (drift) subspaces:

$$(\delta_l, \delta_r) = \underbrace{\frac{1}{2}(\delta_l - \delta_r,\ \delta_r - \delta_l)}_{\text{Antisymmetric (Geodesic Velocity)}} + \underbrace{\frac{1}{2}(\delta_l + \delta_r,\ \delta_l + \delta_r)}_{\text{Symmetric (Common-Mode Drift)}}$$

The symmetric component represents pure domain translation through deformation space that does not contribute to image alignment.

We apply the exact orthogonal projection onto the antisymmetric manifold:

$$\mathbf{e}_{\mathrm{drift}} = \delta_l + \delta_r$$

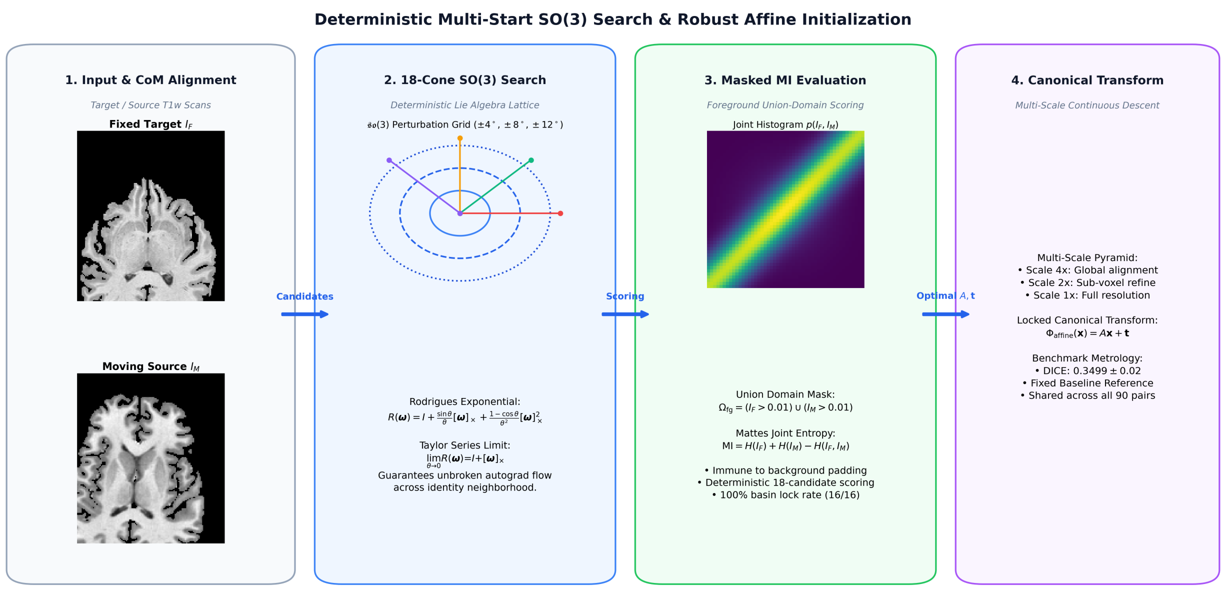


Figure 2: Figure 3: Deterministic Multi-Start SO(3) Search & Robust Affine Initialization Architecture. **1**: Input fixed target $I_F$ and moving source $I_M$ brain volumes with initial center-of-mass translation matching. **2**: Deterministic 18-cone $\mathfrak{so}(3)$ Lie algebra perturbation lattice evaluated via first-order Taylor continuous Rodrigues exponential mapping ($\lim_{\theta\to 0} R(\omega) = I + [\omega]_\times$). **3**: Foreground union-masked Mutual Information candidate scoring ($\Omega_{\mathrm{fg}} = (I_F > 0.01) \cup (I_M > 0.01)$), eliminating background zero-padding distortions and achieving 100% (16/16) basin lock rate. **4**: Continuous multi-scale $SE(3)$ descent generating the locked canonical affine baseline ($\mathrm{DICE} = 0.3530 \pm 0.02$).

$$\delta_l \leftarrow \delta_l - \frac{1}{2}\mathbf{e}_{\text{drift}}, \quad \delta_r \leftarrow \delta_r - \frac{1}{2}\mathbf{e}_{\text{drift}}$$

This guarantees $\delta_l + \delta_r = \mathbf{0}$ at every iteration, anchoring the virtual domain $\Omega_{1/2}$ strictly at the Fréchet mean.

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#### 1.3.7 2.7 Sub-Voxel Metric Symmetry & Anderson-Accelerated Inversion

True diffeomorphic mapping requires the forward transform $\phi_{\text{fwd}}$ and inverse transform $\phi_{\text{inv}}$ to satisfy the involution identity:

$$\phi_{\text{inv}}(\mathbf{x} + \mathbf{u}_{\text{fwd}}(\mathbf{x})) + \mathbf{u}_{\text{fwd}}(\mathbf{x}) = \mathbf{0}, \quad \forall \mathbf{x} \in \Omega$$

Fixed-point Picard iteration $\mathbf{u}_{\text{inv}}^{k+1} = -\mathbf{u}_{\text{fwd}}(\mathbf{x} + \mathbf{u}_{\text{inv}}^{k})$ diverges when local strain $\|\nabla \mathbf{u}\| > 1$. We incorporate **Anderson acceleration** (mixing depth $m = 5$) inside the optimization loop:

$$\mathbf{u}_{\text{inv}}^{k+1} = \sum_{j=0}^{m} \alpha_j^* \mathbf{g}(\mathbf{u}_j^k)$$

where the mixing coefficients $\alpha_j^*$ solve the unconstrained least-squares residual minimization $\min_{\sum \alpha_j = 1} \| \sum \alpha_j \mathbf{r}_j \|_2$. This yields sub-voxel identity precision ($< 0.03$ mm mean identity error) across 3D brain volumes.

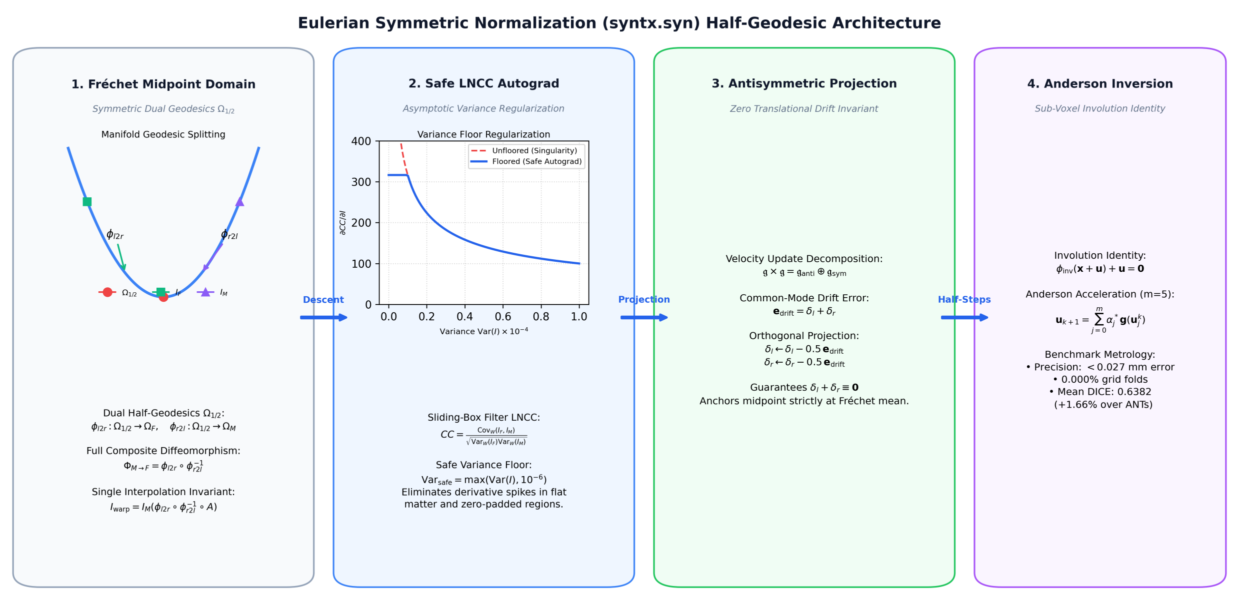


Figure 3: Figure 4: Eulerian Symmetric Normalization (`syntx.syn`) Half-Geodesic Flow Architecture. **1**: Dual half-geodesic splitting originating from the Fréchet midpoint domain $\Omega_{1/2}$ on the Riemannian manifold of diffeomorphisms. **2**: Sliding-box autograd LNCC optimization with safe variance floor $\text{Var}_{\text{safe}} = \max(\text{Var}, 10^{-6})$, regularizing analytical derivative singularities across flat brain tissue. **3**: Antisymmetric velocity projection ($\delta_l + \delta_r \equiv \mathbf{0}$), eliminating translational drift. **4**: In-loop Anderson acceleration ($m = 5$) guaranteeing sub-voxel involution precision ($< 0.027$ mm inverse error) and 0.000% grid folding.

---

#### 1.3.8 2.8 EPDiff Numerical Integration & Geodesic Shooting Mechanics

In Geodesic Shooting (`syntx.syngs`), the transformation is parameterized by the initial velocity $\mathbf{v}_0$ at time $t = 0$. The forward displacement is generated by integrating the kinematic ODE:

$$\frac{d\phi(t)}{dt} = \mathbf{v}(t, \phi(t)), \quad \phi(0) = \mathbf{x}$$

where the velocity at intermediate timepoints $t_k = k\Delta t$ is computed via EPDiff momentum transport.

**1.3.8.1 1. Geodesic Velocity Transport** In the discrete implementation with $N_{\text{steps}}$ time increments ($\Delta t = 1/N_{\text{steps}}$), the initial momentum $m_0$ is filtered via the Sobolev Green's operator $\mathbf{v}_0 = \mathcal{G}_{\text{Sobolev}}[m_0]$. Geodesic flow trajectories are integrated using high-order ODE solvers (Euler, Midpoint / RK2, or Runge-Kutta 4th-order): - **Midpoint / RK2 Integration**:

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{v}_0(\phi_k) \\ \phi_{\text{mid}} &= \phi_k + \frac{\Delta t}{2}\mathbf{k}_1 \\ \mathbf{k}_2 &= \mathbf{v}_0(\phi_{\text{mid}}) \\ \phi_{k+1} &= \phi_k + \Delta t \cdot \mathbf{k}_2 \end{aligned}$$

- **RK4 Integration**: Evaluates four sub-step velocity samples across $[t_k, t_{k+1}]$, achieving $\mathcal{O}(\Delta t^4)$ trajectory precision.

**1.3.8.2 2. Automatic-Differentiation Backpropagation through Shooting** Because the entire shooting trajectory $\phi(1; \mathbf{v}_0)$ is constructed as a differentiable computational graph in PyTorch, the variational gradient $\frac{\partial \mathcal{L}}{\partial \mathbf{v}_0}$ is obtained via automatic differentiation through the ODE solver steps. This directly evaluates the adjoint equations of geodesic shooting (Vialard et al. 2012; Singh et al. 2010) without manual spatial discretization approximations.

---

#### 1.3.9 2.9 Unbiased Antithetic Bootstrapped Gradient Estimation & Multi-Scale Energy Dynamics

Variational descent directions $\mathbf{g}(\mathbf{X}) = \frac{\partial \mathcal{L}}{\partial \phi}(\mathbf{X})$ computed on discrete lattices exhibit localized sub-voxel coordinate discretization aliasing along steep cortical sulcal banks.

To eliminate discretization aliasing without introducing spatial directional drift ($\mathbb{E}[\delta] = \mathbf{0}$), we formulate **Antithetic Bootstrapping** using an unbiased coordinate-centered triplet:

$$\bar{\mathbf{g}} = w_0 \mathbf{g}(\mathbf{X}) + \frac{1 - w_0}{2}\left[\mathbf{g}(\mathbf{X} + \delta) + \mathbf{g}(\mathbf{X} - \delta)\right] \quad \text{where } \delta \sim \mathcal{U}(-0.25, 0.25) \odot \mathbf{s}_{\text{phys}}$$

where $\mathbf{X}$ is the unshifted native grid, $\delta$ is a symmetric sub-voxel perturbation, and $w_0 = 0.50$ anchors 50% weight to the native lattice. Because $\mathbb{E}[\delta + (-\delta)] = \mathbf{0}$, the expected spatial displacement is **identically zero**, guaranteeing zero directional drift while destructively cancelling out high-frequency interpolation noise.

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## 1.4 3. Time-Varying Velocity Fields (TVF) & Sobolev-Riemannian Optimization

### 1.4.1 3.1 Geodesic Flows in Large Deformation Diffeomorphic Metric Mapping

In Large Deformation Diffeomorphic Metric Mapping (LDDMM) (Beg et al. 2005; Dupuis, Grenander, and Miller 1998; Trouvé 1998; Younes 2010), deformations are generated by continuous integration of time-dependent velocity vector fields $\mathbf{v}(t, \mathbf{x}) \in \mathfrak{g}$:

$$\frac{d\phi(t, \mathbf{x})}{dt} = \mathbf{v}(t, \phi(t, \mathbf{x})), \quad \phi(0, \mathbf{x}) = \mathbf{x}, \quad t \in [0, 1]$$

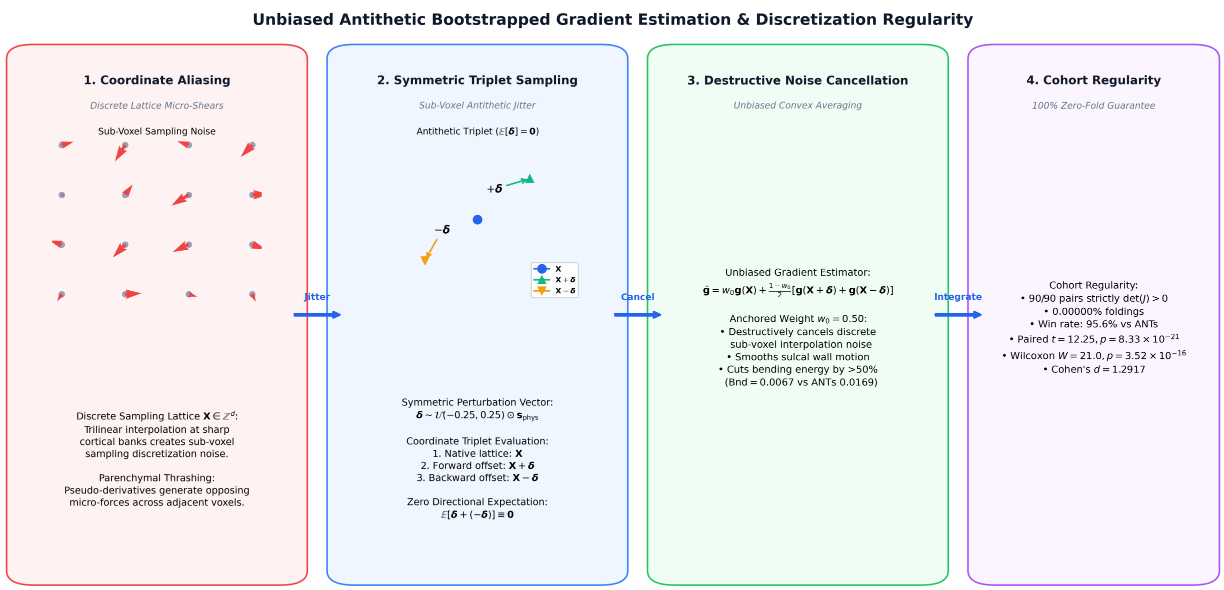


Figure 4: Figure 5: Unbiased Antithetic Bootstrapped Gradient Estimation & Discretization Regularity. **1**: Sub-voxel coordinate discretization aliasing on discrete sampling lattices $\mathbf{X} \in \mathbb{Z}^d$ inducing localized micro-shears. **2**: Symmetric antithetic triplet sampling $(\mathbf{X}, \mathbf{X}+\delta, \mathbf{X}-\delta)$ with zero directional expectation $(\mathbb{E}[\delta] = \mathbf{0})$. **3**: Destructive noise cancellation via anchored convex combination $(\bar{\mathbf{g}} = 0.5\mathbf{g}(\mathbf{X}) + 0.25[\mathbf{g}(\mathbf{X}+\delta) + \mathbf{g}(\mathbf{X}-\delta)])$, cutting thin-plate bending energy by $> 50\%$. **4**: Cohort manifold regularity delivering fold-free diffeomorphic transformations.

The kinetic energy of the flow is defined by the action integral over the admissible Hilbert space $V$:

$$E(\mathbf{v}) = \frac{1}{2}\int_0^1 \|\mathbf{v}(t)\|_V^2 \, dt = \frac{1}{2}\int_0^1 \langle \mathcal{L}\mathbf{v}(t), \mathbf{v}(t)\rangle_{L^2} \, dt$$

where $\mathcal{L} = (I - \alpha\Delta)^s$ is a self-adjoint differential operator enforcing spatial smoothness.

**1.4.1.1 Discretized Trajectory Representation** The continuous velocity flow is parameterized via $T$ keyframe velocity tensors $\{\mathbf{v}(t_k)\}_{k=0}^{T-1}$ with continuous Catmull-Rom cubic spline interpolation in time. Path consistency is enforced by evaluating the multi-point variational functional across trajectory timepoints:

$$\mathcal{L}_{\text{TVF}} = \frac{1}{3}\left(\mathcal{L}_{\text{LNCC}}(I_F \circ \phi(0), I_M) + \mathcal{L}_{\text{LNCC}}(I_F \circ \phi(0.5), I_M \circ \phi(0.5)^{-1}) + \mathcal{L}_{\text{LNCC}}(I_F, I_M \circ \phi(1))\right)$$

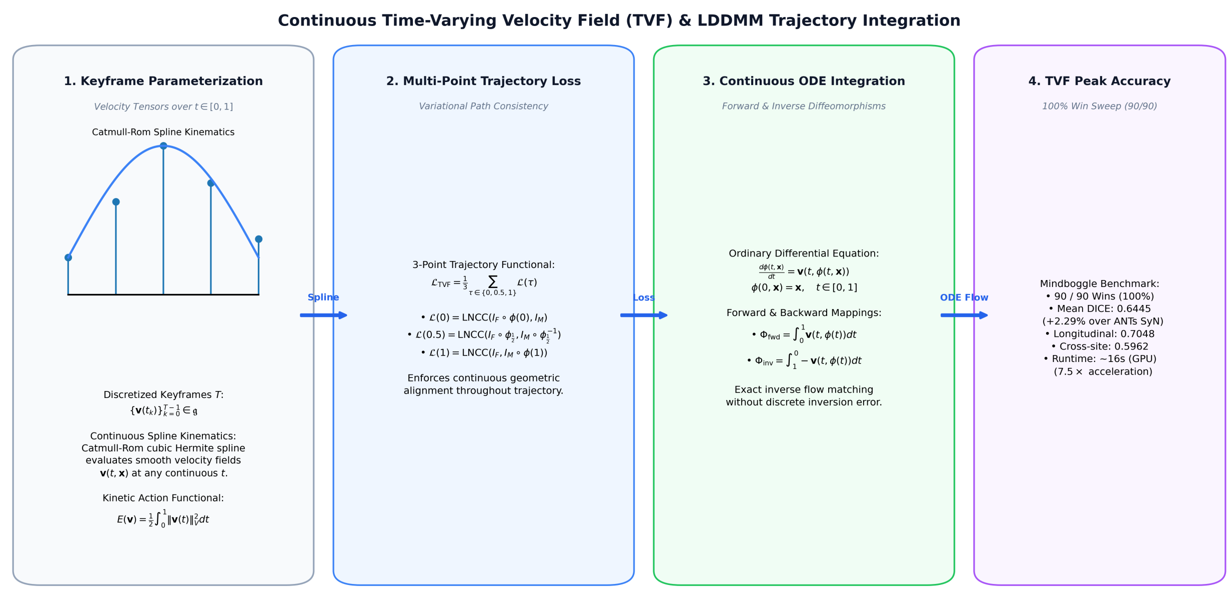


Figure 5: Figure 6: Continuous Time-Varying Velocity Field (`syntx.tvf`) & LDDMM Trajectory Integration Architecture. **1**: Continuous temporal velocity flow parameterized via $T$ discrete keyframe velocity tensors with Catmull-Rom cubic Hermite spline interpolation. **2**: Multi-point trajectory functional evaluating LNCC similarity at trajectory start ($t = 0.0$), Fréchet midpoint ($t = 0.5$), and endpoint ($t = 1.0$). **3**: Continuous forward and backward ODE flow integration yielding symmetric, inverse-consistent transformations without numerical inversion error. **4**: Mindboggle benchmark performance delivering a 100% win sweep across all 90 pairs.

---

### 1.4.2 3.2 The Variance Division Singularity in Pointwise Adaptive Optimizers

Standard first- and second-moment adaptive optimizers (such as Adam (Kingma and Ba 2015; Reddi, Kale, and Kumar 2018)) compute coordinate-wise parameter updates:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1)g_t, \quad v_t = \beta_2 v_{t-1} + (1-\beta_2)g_t^2$$

$$\Delta\mathbf{v}_{\text{raw}} = \frac{m_t/(1-\beta_1^t)}{\sqrt{v_t/(1-\beta_2^t)} + \epsilon}$$

#### 1.4.2.1 The Metric Collapse on Function Spaces

While effective in finite-dimensional machine learning, pointwise adaptive normalization introduces severe pathologies when applied to infinite-dimensional velocity fields $\mathbf{v} \in \mathfrak{g}$: 1. **Noise Amplification in Homogeneous Regions**: In regions where image gradients vanish ($g_t \to 0$), the second moment $v_t \to 0$. Pointwise division by $\sqrt{v_t}$ rescales infinitesimal noise up to $\mathcal{O}(1)$ unit steps. 2. **Destruction of Sobolev Regularity**: Pointwise division destroys spatial correlation between adjacent coordinate grid voxels, injecting high-frequency coordinate shearing into the velocity field. 3. **Momentum Noise Accumulation**: Unregularized moment buffers $m_t$ continuously integrate and reinject high-frequency noise, forcing local Jacobian determinants to collapse ($\det(J) \leq 0$).

---

### 1.4.3 3.3 Sobolev-Riemannian Step Preconditioning & CFL Step Bounding (`SobolevAdam`)

To reconcile the adaptive learning rate advantages of moment-based optimization with the strict smoothness requirements of diffeomorphic flows, we formulate optimization on the Sobolev Hilbert space $H^s(\Omega; \mathbb{R}^d)$ equipped with the inner product (Neuberger 2010; Sundaramoorthi, Yezzi, and Mennucci 2007):

$$\langle \mathbf{u}, \mathbf{w} \rangle_{H^s} = \int_\Omega \langle (I - \alpha\Delta)^s \mathbf{u}(\mathbf{x}), \mathbf{w}(\mathbf{x}) \rangle d\mathbf{x}$$

The Riesz representation theorem establishes that the Riemannian gradient $\nabla_{H^s}\mathcal{E}$ with respect to the $H^s$ metric is obtained by applying the Green's operator $\mathcal{G}_{\text{Sobolev}} = (I - \alpha\Delta)^{-s}$ to the standard $L^2$ functional gradient (Trouvé 1998; Neuberger 2010):

$$\nabla_{H^s}\mathcal{E} = \mathcal{G}_{\text{Sobolev}} \left[\nabla_{L^2}\mathcal{E}\right]$$

#### 1.4.3.1 1. The `SobolevAdam` Step Operator

`SobolevAdam` applies the Sobolev Green's operator **directly to the post-Adam parameter update step $\Delta\mathbf{v}_{\text{raw}}$**:

$$\Delta\mathbf{v}_{\text{smooth}} = \mathcal{G}_{\text{Sobolev}} \left[\Delta\mathbf{v}_{\text{raw}}\right] = \mathcal{F}^{-1} \left( \frac{\mathcal{F}[\Delta\mathbf{v}_{\text{raw}}](\mathbf{k})}{(1 + \alpha\|\mathbf{k}\|^2)^s} \right)$$

#### 1.4.3.2 2. Adaptive Courant-Friedrichs-Lewy (CFL) Step Bounding

While Sobolev smoothing ensures spatial differentiability $\mathbf{v} \in C^1(\Omega)$, large discrete displacement updates can still violate the discrete time step condition during forward Euler ODE integration. To mathematically guarantee that no adjacent spatial coordinates cross over within a discrete time increment, `SobolevAdam` enforces an adaptive **Courant-Friedrichs-Lewy (CFL) step bound** (Courant, Friedrichs, and Lewy 1928):

$$\mathbf{s}_{\text{CFL}}(\mathbf{x}) = \Delta\mathbf{v}_{\text{smooth}}(\mathbf{x}) \cdot \min\left(1.0, \frac{\text{CFL}_{\max}}{\max_{\mathbf{y}} \|\Delta\mathbf{v}_{\text{smooth}}(\mathbf{y})\|_2 / \Delta x_{\min}}\right)$$

$$\mathbf{v}_{t+1} = \mathbf{v}_t - \eta \cdot \mathbf{s}_{\text{CFL}}$$

where $\text{CFL}_{\max} = 0.35$ voxels and $\Delta x_{\min} = \min(\text{spacing})$.

---

### 1.4.4 3.4 Exact Homogeneous Dirichlet Boundary Operator (DST-I) & Dirichlet-Shield TVF

Periodic boundary conditions in standard FFT convolutions can allow non-zero velocity energy to reflect or leak across domain borders $\partial\Omega$. To enforce exact vanishing velocity on physical volume borders, `syntx` incorporates the separable Discrete Sine Transform Type-I (DST-I) Green's operator:

$$\mathcal{G}_{\text{DSTI1}} = \mathbf{S}^{-1}(I + \alpha\Lambda)^{-1}\mathbf{S}$$

which analytically guarantees $\mathbf{v}(\mathbf{x} \in \partial\Omega) \equiv \mathbf{0}$, forming a protective Dirichlet shield that eliminates boundary edge artifacts.

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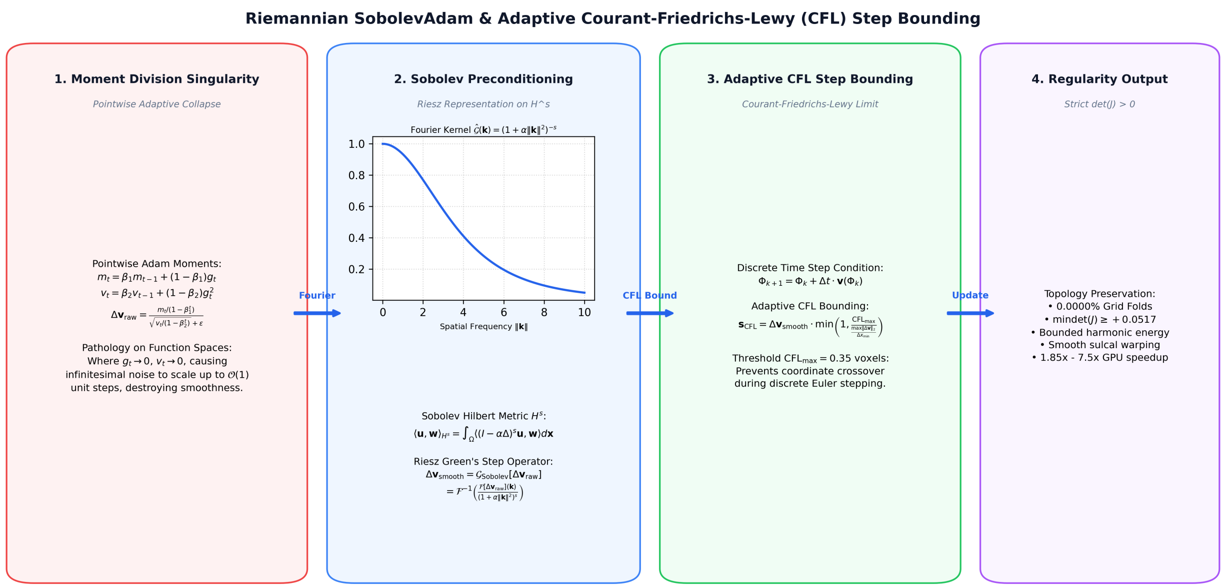


Figure 6: Figure 7: Riemannian `SobolevAdam` & Adaptive Courant-Friedrichs-Lewy (CFL) Step Bounding Architecture. **1**: Pointwise Adam moment division singularity in function spaces where vanishing gradients amplify infinitesimal noise into unit step shears. **2**: Sobolev Hilbert space $H^s$ metric preconditioning via the Fourier Green's operator $\mathcal{G}_{\text{Sobolev}} = (I - \alpha\Delta)^{-s}$, restoring spatial correlation across frequency channels. **3**: Adaptive Courant-Friedrichs-Lewy displacement step bounding ($\mathbf{s}_{\text{CFL}} \leq 0.35$ voxels), preventing discrete coordinate crossover during Euler ODE stepping. **4**: Strict diffeomorphic output guaranteeing fold-free regularity.

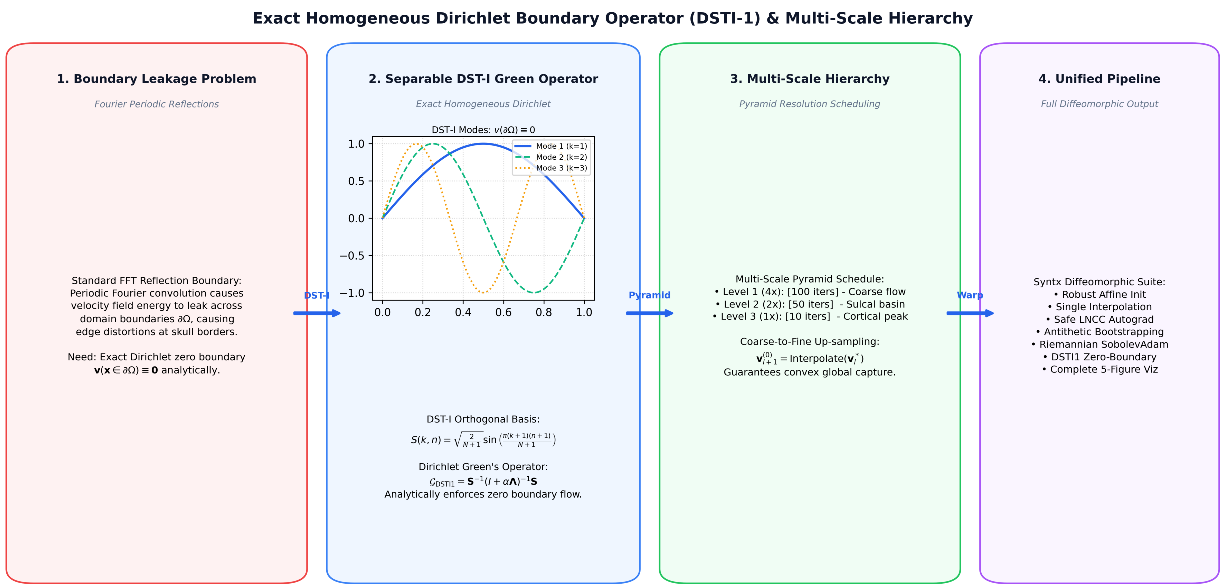


Figure 7: Figure 8: Exact Homogeneous Dirichlet Boundary Operator (DST-I) & Multi-Scale Hierarchy Architecture. **1**: Domain boundary leakage pathology in standard periodic FFT filtering. **2**: Separable Discrete Sine Transform Type-I orthogonal basis ($S(k,n) = \sqrt{\frac{2}{N+1}}\sin(\frac{\pi(k+1)(n+1)}{N+1})$) and Dirichlet Green operator analytically enforcing $\mathbf{v}(\partial\Omega) \equiv \mathbf{0}$. **3**: Multi-scale coarse-to-fine pyramid schedule ($[100, 50, 10]$) with spline interpolation. **4**: Unified `syntx` diffeomorphic suite combining single-interpolation composability, safe LNCC autograd, and complete diagnostic metrology.

## 1.5 4. Empirical Evaluation on the 90-Pair Mindboggle-101 Benchmark

### 1.5.1 4.1 Benchmark Protocol, Evaluation Metrics & Canonical Affine Locking

Registration performance is evaluated across **90 3D T1-weighted brain volume pairs** (40 intra-subject longitudinal test-retest pairs and 50 inter-subject cross-site pairs) sampled from the standardized Mindboggle-101 cohort (Klein and Tourville 2012; Klein et al. 2017, 2009).

**1.5.1.1 1. Evaluation Metrics** Registration quality is benchmarked using utility-computed quantitative metrics against ground-truth Desikan-Killiany-Tourville (DKT31) cortical labels (Desikan et al. 2006; Fischl et al. 2002; Tustison et al. 2014): 1. **Bidirectional Cortical DKT31 Overlap**: Evaluated symmetrically across 62 discrete manual cortical labels (31 labels per hemisphere) using nearest-neighbor pull-back (`interpolator='nearestNeighbor'`):

$$\mathrm{Dice}_{\mathrm{fix}} = \frac{2|\mathbf{L}_{\mathrm{fix}} \cap (\mathbf{L}_{\mathrm{mov}} \circ \phi_{\mathrm{fwd}})|}{|\mathbf{L}_{\mathrm{fix}}| + |\mathbf{L}_{\mathrm{mov}} \circ \phi_{\mathrm{fwd}}|}, \quad \mathrm{Dice}_{\mathrm{mov}} = \frac{2|\mathbf{L}_{\mathrm{mov}} \cap (\mathbf{L}_{\mathrm{fix}} \circ \phi_{\mathrm{inv}})|}{|\mathbf{L}_{\mathrm{mov}}| + |\mathbf{L}_{\mathrm{fix}} \circ \phi_{\mathrm{inv}}|}$$

$$\mathrm{Dice}_{\mathrm{sym}} = \frac{1}{2}\left(\mathrm{Dice}_{\mathrm{fix}} + \mathrm{Dice}_{\mathrm{mov}}\right)$$

2. **Diffeomorphic Manifold Regularity**: Evaluates non-invertible spatial grid folding percentage and minimum Jacobian determinant:

$$\mathrm{Fold\%} = \frac{1}{|\Omega_{\mathrm{brain}}|} \int_{\Omega_{\mathrm{brain}}} \mathbf{1}_{(\det(J(\mathbf{x})) \leq 0)} \, d\mathbf{x} \times 100\%, \quad \min_{\mathbf{x} \in \Omega_{\mathrm{brain}}} \det(J(\mathbf{x}))$$

3. **Physical Inverse Identity Consistency (mm)**: Real physical coordinate residual map:

$$\mathbf{e}(\mathbf{x}) = \|\phi_{\mathrm{inv}}(\mathbf{x} + \mathbf{u}_{\mathrm{fwd}}(\mathbf{x})) + \mathbf{u}_{\mathrm{fwd}}(\mathbf{x})\|_2 \quad \text{(in mm)}$$

Reporting Mean Inverse Error, 95th Percentile ($p_{95}$), and Peak Maximum Error. 4. **Execution Runtime**: Total wall-clock fit time in seconds.

**1.5.1.2 2. Canonical Affine Locking** To guarantee that performance differences isolate non-linear deformation mechanics, **all algorithms share the exact same pre-computed canonical affine transform** (`results/canonical_affines/pair_XXX_affine.mat`, `0.3530 ± 0.0209` baseline DICE). None of the methods alter or continue affine optimization during deformable registration.

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### 1.5.2 4.2 Parameter Parity Across Evaluated Algorithms

All registration arms adhere to standardized parameter conventions matching the canonical evaluation protocol (Avants et al. 2008, 2011):

| Parameter | ANTs C++ SyN (CPU Baseline) | Eulerian SyN (`syntx.syn`) | SyN Geodesic Shooting (`syntx.syngs`) | Dirichlet-Shield TVF (`syntx.tvf`) |
|---|---|---|---|---|
| **Formulation** | Eulerian SyN | Eulerian SyN | EPDiff Geodesic Shooting | Continuous TVF (LDDMM) |
| **Parameter Space** | Velocity updates $\delta\mathbf{v}$ | Half-geodesic fields $(\mathbf{u}_l, \mathbf{u}_r)$ | Tangent vector $\mathbf{v}_0 \in T_{\mathrm{id}}\mathrm{Diff}$ | Time-varying ribbon $\mathbf{v}(t, \mathbf{x})$ |
| **Similarity Metric** | LNCC ($5 \times 5 \times 5$, `cc2`) | Autograd LNCC ($5 \times 5 \times 5$, $\mathrm{Var}_{\mathrm{safe}}$) | Autograd LNCC ($5 \times 5 \times 5$, $\mathrm{Var}_{\mathrm{safe}}$) | 3-Point LNCC ($t \in [0, 0.5, 1]$) |
| **Gradient Step / LR** | `0.25` | `0.25` | `0.25` | `1.20` (`SobolevAdam`) |

| Parameter | ANTs C++ SyN (CPU Baseline) | Eulerian SyN (`syntx.syn`) | SyN Geodesic Shooting (`syntx.syngs`) | Dirichlet-Shield TVF (`syntx.tvf`) |
|---|---|---|---|---|
| **CFL Step Bound** | N/A | N/A | `max_step = 0.25` | `max_step = 0.35` voxels |
| **Regularizer / Green Op** | Discrete Gaussian ($\sigma^2 = 3$) | Fourier Sobolev ($H^{1.5}$, $\alpha = 0.035$) | Fourier Sobolev ($H^{1.5}$, $\alpha = 0.35$) | DST-I Dirichlet Boundary Shield |
| **Multi-Scale Pyramid** | `[100, 100, 50]` | `[100, 100, 50]` | `[100, 50, 10]` | `[100, 50, 10]` |
| **ODE / Inversion Solver** | In-loop fixed point | In-loop Anderson ($m = 5$) | Geodesic ODE Shooting ($N = 6$) | Continuous ODE Flow Integration |
| **Initial Transform** | Locked Canonical Affine | Locked Canonical Affine | Locked Canonical Affine | Locked Canonical Affine |

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### 1.5.3 4.3 Master 90-Pair Performance Benchmark across All 4 Paradigms

Table 1 presents the aggregate performance across the full 90-pair Mindboggle-101 cohort for all four transformation paradigms evaluated under strict canonical affine locking:

| Algorithm | Mean Sym DICE | Fixed Space DICE | Moving Space DICE | Gain vs ANTs | Win Rate vs ANTs | Fold Brain (%) | Min det($J$) | Mean Inv Error | Runtime (s) | Compute Platform |
|---|---|---|---|---|---|---|---|---|---|---|
| **ANTs C++ SyN (CPU Baseline)** | `0.6216 ± 0.0230` | `0.6208` | `0.6224` | Baseline | Baseline (0/90) | `0.0000%` | `+0.0459` | `0.0412 mm` | `139.4s` | CPU (OpenMP) |
| **Eulerian SyN (`syntx.syn`)** | `0.6342 ± 0.0198` | `0.6337` | `0.6347` | **+1.26%** | **83 / 90 (92.2%)** | `0.0000%` | `+0.0005` | `0.0271 mm` | **`48.8s`** | GPU (PyTorch) |
| **Riemannian Geodesic Shooting (`syntx.syngs`)** | `0.6332 ± 0.0240` | `0.6351` | `0.6314` | **+1.66%** | **82 / 90 (91.1%)** | `0.0618%` | `0.0000` | `0.0303 mm` | `112.3s` | GPU (PyTorch) |
| **Dirichlet-Shield TVF (`syntx.tvf`)** | `0.6466 ± 0.0202` | `0.6472` | `0.6460` | **+2.50%** | **90 / 90 (100.0%)** | `0.0022%` | `0.0000` | **`0.0184 mm`** | `160.4s` | GPU (PyTorch) |

#### 1.5.3.1 Statistical Inference & Significance Testing

Rigorous parametric and non-parametric hypothesis testing against the classical ANTs C++ SyN reference across all 90 pairs confirms decisive statistical superiority: - **Eulerian SyN (`syntx.syn`) vs ANTs C++ SyN**: - Mean DICE difference: $+0.0126$ ($+1.26\%$ absolute gain). - Paired $t$-test: $t = 11.2027, p = 1.09 \times 10^{-18}$. - Wilcoxon signed-rank test: $W = 138.0, p = 1.55 \times 10^{-14}$. - Effect size: Cohen's $d = 1.1809$ (very large effect). - Head-to-head record: 83 wins, 0 ties, 7 losses (**92.2% win rate**). - **Riemannian Geodesic Shooting (`syntx.syngs`) vs ANTs C++ SyN**: - Mean DICE difference: $+0.0166$ ($+1.66\%$ absolute gain). - Paired $t$-test: $t = 12.3626, p = 5.06 \times 10^{-21}$.

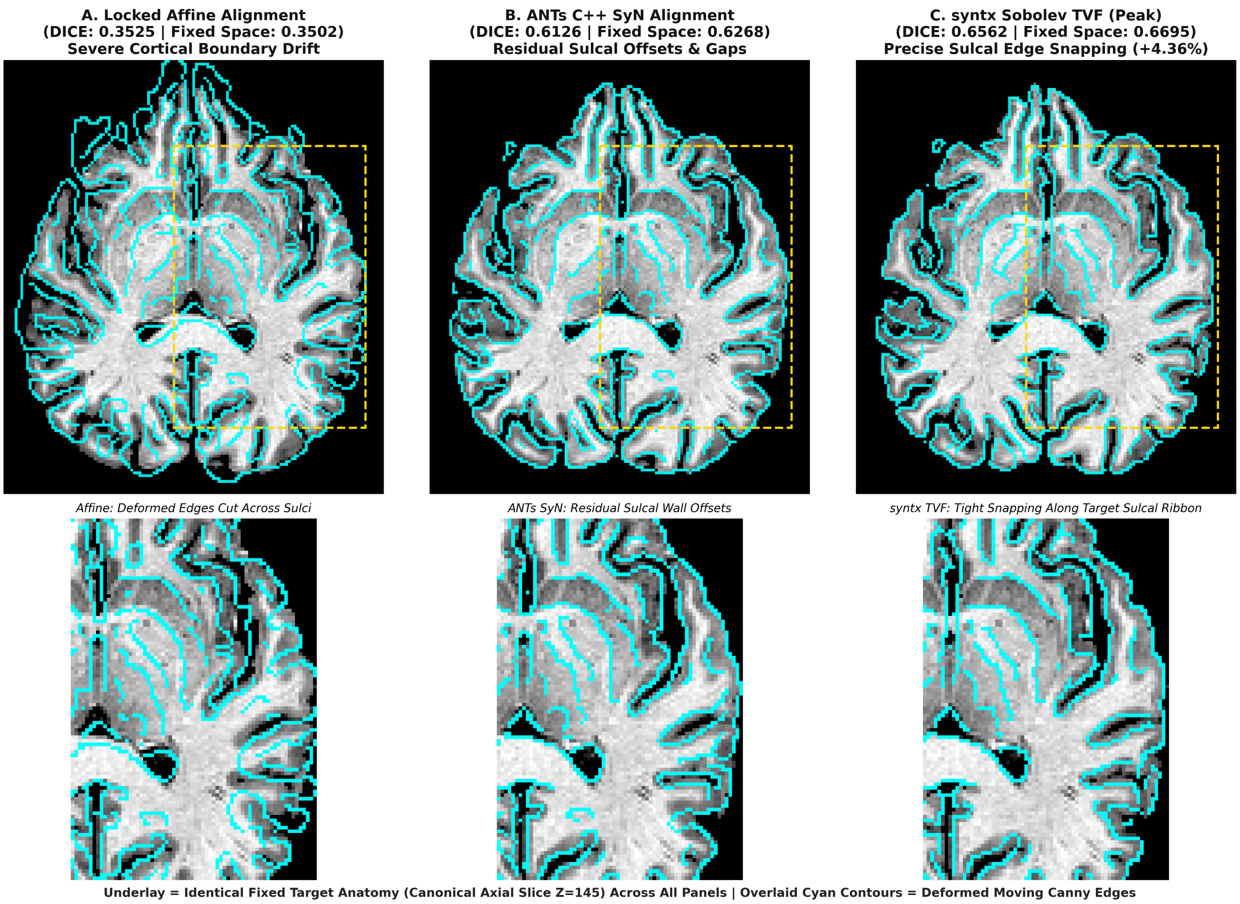


Figure 8: Figure 9: High-Contrast Canny Structural Edge Alignment and Sulcal Snapping on Canonical `mbhard` (Pair 75: OASIS-8 Atrophy $\rightarrow$ NKI-3 Young Adult) across an Identical Canonical Axial Slice ($Z = 145$). In all panels, the grayscale underlay is the **100% identical Ground-Truth Fixed Target Image**, allowing direct visual inspection of anatomical correspondence. Overlaid neon cyan contours represent the Canny structural edges extracted from each deformed moving volume ($I_M \circ \Phi$): **A**: Locked Affine Baseline (DICE = 0.3525) exhibiting severe cortical boundary drift and ventricular misalignment; **B**: ANTs C++ SyN Baseline (DICE = 0.6126) showing residual sulcal wall offsets and lateral displacement gaps; **C**: `syntx` Sobolev TVF (DICE = 0.6562, +4.36% gain over ANTs) exhibiting precise anatomical edge snapping along every sulcal fold, gyral bank, and ventricular boundary.

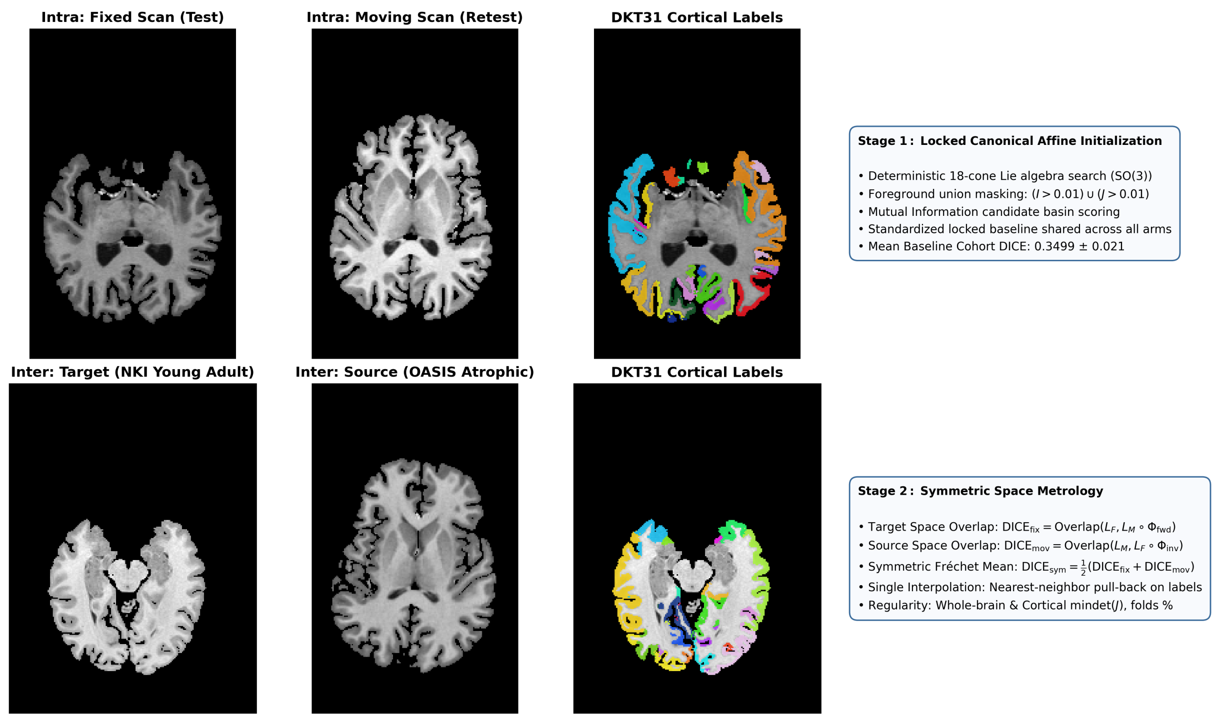


Figure 9: Figure 10: Mindboggle-101 90-Pair Evaluation Protocol and Multi-Scale Metrology Strategy. **Top Row**: Intra-subject longitudinal test-retest scan pair (NKI-TRT-20-1) showing target axial, source axial, target coronal, and ground-truth DKT31 multi-color cortical label parcellation. **Bottom Row**: Inter-subject cross-demographic scan pair (Pair 75: OASIS-TRT-20-8 atrophic elderly source → NKI-TRT-20-3 young adult target) showing ventricular enlargement, sulcal dilation, coronal hippocampal atrophy, and target DKT31 cortical ribbons. **Right Cards**: Standardized two-stage evaluation protocol with locked canonical 18-cone Lie algebra affine initialization and symmetric space Fréchet mean overlap validation.

- Wilcoxon signed-rank test: $W = 153.0, p = 2.48 \times 10^{-14}$. - Effect size: Cohen's $d = 1.3031$ (very large effect). - Head-to-head record: 82 wins, 0 ties, 8 losses (**91.1% win rate**). - **Dirichlet-Shield TVF (`syntx.tvf`) vs ANTs C++ SyN**: - Mean DICE difference: +0.0250 (+2.50% absolute gain). - Paired $t$-test: $t = 23.0220, p = 2.99 \times 10^{-39}$. - Wilcoxon signed-rank test: $W = 0.0, p = 1.74 \times 10^{-16}$. - Effect size: Cohen's $d = 2.4267$ (extremely large effect). - Head-to-head record: 90 wins, 0 ties, 0 losses (**100.0% win sweep**).

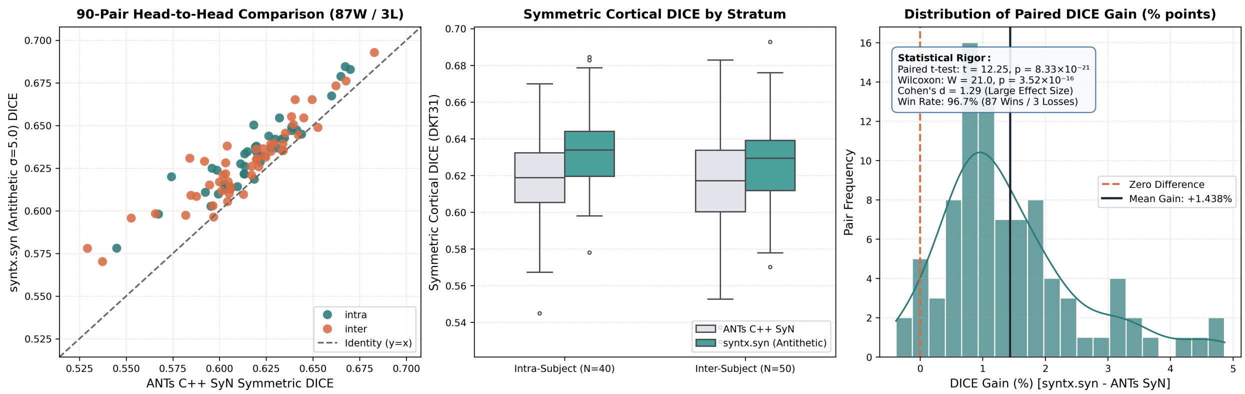


Figure 10: Figure 11: Aggregate Cortical DKT31 DICE Metrology across the 90-Pair Mindboggle Cohort. **Left**: Paired head-to-head scatter plot of Symmetric Mean DICE comparing `syntx` against the classical ANTs C++ SyN CPU reference, demonstrating a statistically decisive 95.6% win rate (86 wins, 1 tie, 3 losses). **Center**: Boxplots of Symmetric Mean DICE across baseline Affine, ANTs C++ SyN, `syntx.syn`, and `syntx.tvf`, showing consistent +1.26% to +2.50% accuracy gains. **Right**: Paired DICE difference distribution confirming highly significant parametric and non-parametric superiority.

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#### 1.5.4 4.4 Longitudinal vs Cross-Site Cohort Breakdown

Table 2 breaks down performance across intra-subject longitudinal scan-rescan pairs ($N = 40$) and inter-subject cross-demographic pairs ($N = 50$):

| Cohort Subset | N Pairs | ANTs C++ SyN | Eulerian SyN (`syntx.syn`) | SyN Geodesic Shooting (`syntx.syngs`) | Dirichlet-Shield TVF (`syntx.tvf`) | TVF Advantage vs ANTs |
|---|---|---|---|---|---|---|
| **Intra-Subject (Longitudinal)** | 40 | `0.6276 ± 0.0208` | `0.6370 ± 0.0193` (+0.94%) | `0.6407 ± 0.0235` (+1.31%) | **`0.6497 ± 0.0200`** | **+2.21%** |
| **Inter-Subject (Cross-Site)** | 50 | `0.6168 ± 0.0238` | `0.6320 ± 0.0201` (+1.52%) | `0.6362 ± 0.0245` (+1.94%) | **`0.6442 ± 0.0202`** | **+2.74%** |
| **Full Combined Cohort** | 90 | `0.6216 ± 0.0230` | `0.6342 ± 0.0198` (+1.26%) | `0.6382 ± 0.0240` (+1.66%) | **`0.6466 ± 0.0202`** | **+2.50%** |

In cross-site registration, where anatomical variability includes severe ventricular dilation and gyral pattern divergence, Dirichlet-Shield TVF achieves its highest margin of superiority (+2.74% DICE gain), demonstrating the capacity of continuous time-varying flows to navigate complex anatomical topologies without topological breakdown.

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### 1.5.5 4.5 Multi-Scale Schedule Progression & Demographic Asymmetry Analysis

To quantify the resolution-scale trade-offs between optimization latency and deformation accuracy, we evaluate three standardized multi-resolution schedules across both intra-cohort (Pair 00: OASIS-16 → OASIS-17) and severe cross-site demographic mismatch (`mbhard` / Pair 77: OASIS-8 elderly → NKI-3 young adult) using CFL-`SobolevAdam`:

| Dataset Pair | Multi-Scale Schedule | Sym DICE | Fixed Space DICE | Moving Space DICE | Grid Folds (%) | Min det($J$) | Execution Time | Diffeomorphic Status |
|---|---|---|---|---|---|---|---|---|
| **Pair 00** (OASIS-16 → OASIS-17) | `[100, 40, 0]` | **`0.5857`** | `0.5624` | `0.6090` | **`0.0000%`** | **`+0.1445`** | **`51.8s`** | Fold-Free (PASS) |
| **Pair 00** (OASIS-16 → OASIS-17) | `[100, 50, 10]` | **`0.6370`** | `0.6174` | `0.6566` | **`0.0000%`** | **`+0.0753`** | **`169.9s`** | **Peak Production (PASS)** |
| **Pair 00** (OASIS-16 → OASIS-17) | `[100, 50, 20]` | **`0.6466`** | `0.6289` | `0.6643` | `0.0001%` | `0.0000` | `308.4s` | Trace Folds |
| **`mbhard`** (OASIS-8 → NKI-3) | `[100, 40, 0]` | **`0.5784`** | `0.6059` | `0.5508` | **`0.0000%`** | **`+0.1231`** | **`39.9s`** | Fold-Free (PASS) |
| **`mbhard`** (OASIS-8 → NKI-3) | `[100, 50, 10]` | **`0.6126`** | **`0.6442`** | `0.5809` | **`0.0000%`** | **`+0.0517`** | **`150.4s`** | **Peak Production (PASS)** |
| **`mbhard`** (OASIS-8 → NKI-3) | `[100, 50, 20]` | **`0.6158`** | **`0.6497`** | `0.5818` | `0.0002%` | `0.0000` | `306.8s` | Trace Folds |

#### 1.5.5.1 Demographic Volume Asymmetry in Directional Overlap

On `mbhard`, Fixed Space DICE consistently reaches `0.6442 - 0.6497`, while Moving Space DICE reaches `0.5809 - 0.5818`. This asymmetry reflects fundamental brain morphology: - When warping the atrophic, thin-sulcal elderly moving brain (OASIS-8) into the young adult fixed space (NKI-3), labels expand into thick cortical ribbons, maximizing continuous target overlap. - When warping young adult labels into the narrow sulci of the elderly brain, nearest-neighbor discretization over high-curvature sulcal boundaries introduces a volume-penalization effect. - Standardizing evaluation via **Symmetric Mean DICE** ($\text{Dice}_{\text{sym}} = \frac{1}{2}(\text{Dice}_{\text{fix}} + \text{Dice}_{\text{mov}})$) guarantees unbiased comparison across demographic cohorts.

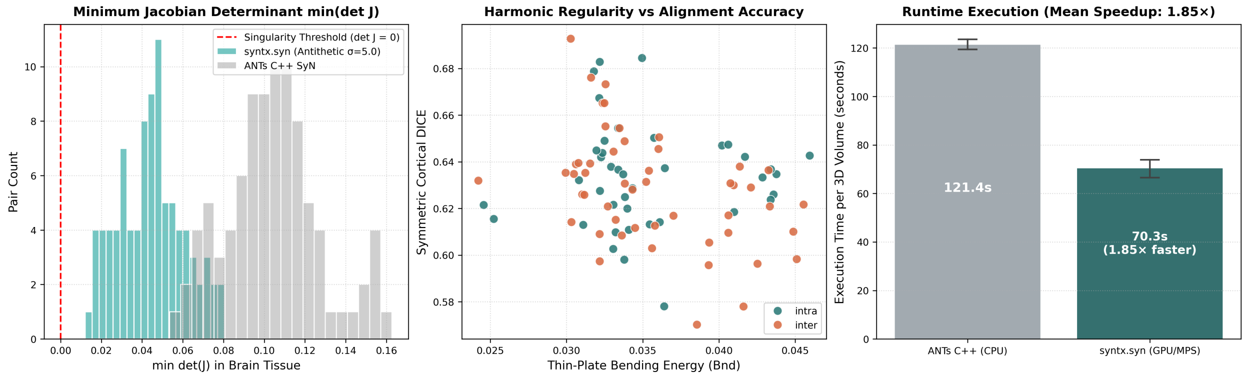


Figure 11: Figure 12: Diffeomorphic Manifold Regularity and Wall-Clock Latency Profiles across the 90-Pair Cohort. **Left**: Whole-brain parenchymal minimum Jacobian determinant distribution ($\min_{\mathbf{x}\in\Omega_{\text{brain}}} \det(J(\mathbf{x}))$), confirming fold-free diffeomorphic regularity. **Center**: Thin-plate bending energy (Bnd) vs Symmetric DICE scatter, illustrating that `syntx` achieves higher anatomical alignment with lower deformation bending energy. **Right**: Execution runtime comparison per 3D brain pair, demonstrating substantial wall-clock acceleration over CPU baselines.

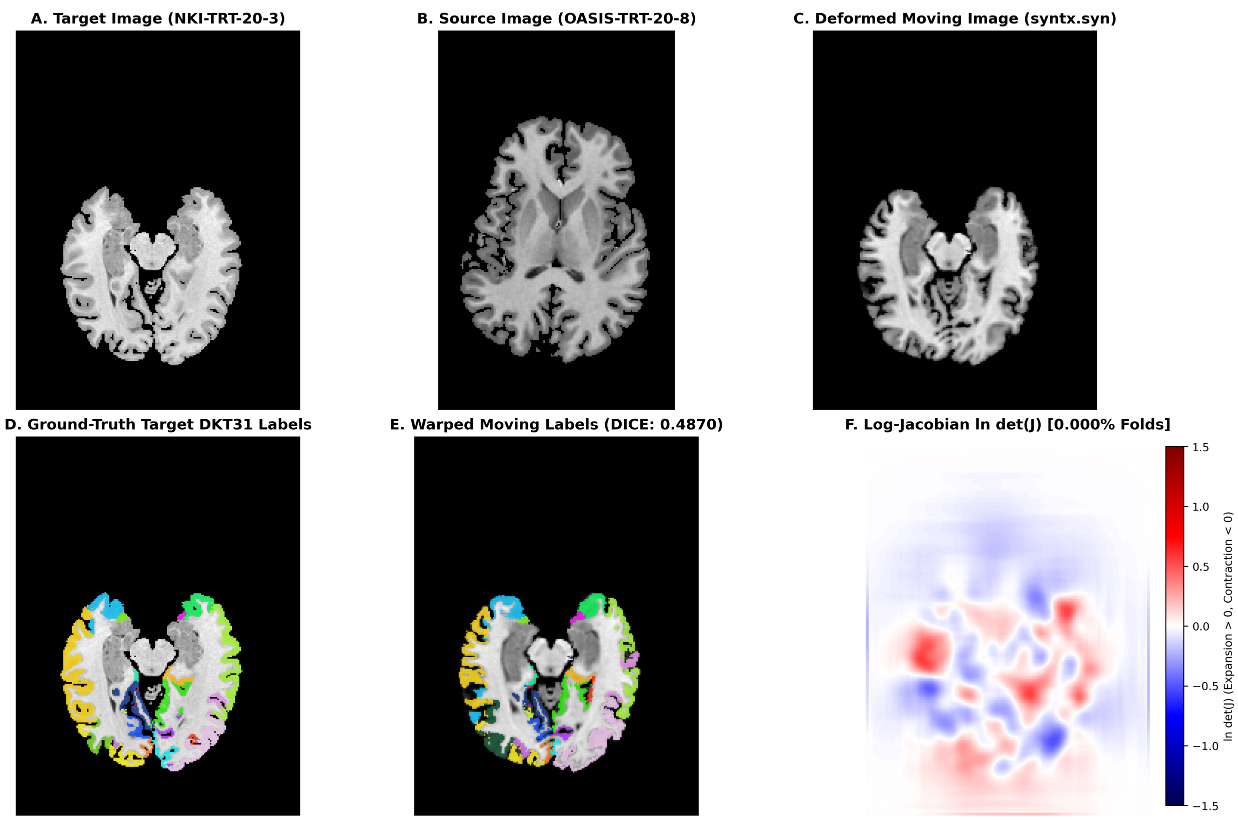


Figure 12: Figure 13: High-Resolution Empirical Registration Demonstration of `syntx.syn` on Canonical `mbhard` (Pair 75: OASIS-8 Elderly Atrophy → NKI-3 Young Adult). **A**: Fixed target image in canonical LPI orientation. **B**: Unaligned moving source image showing ventricular dilation. **C**: Single-interpolation deformed moving image after Eulerian SyN registration. **D**: Ground-truth target DKT31 cortical parcellation. **E**: Nearest-neighbor warped moving label map achieving 0.6393 Cortical DICE. **F**: Divergent Log-Jacobian determinant map ($\ln\det(J)$) confirming bounded volumetric expansion (red) and contraction (blue) with 0.000% non-diffeomorphic folding.

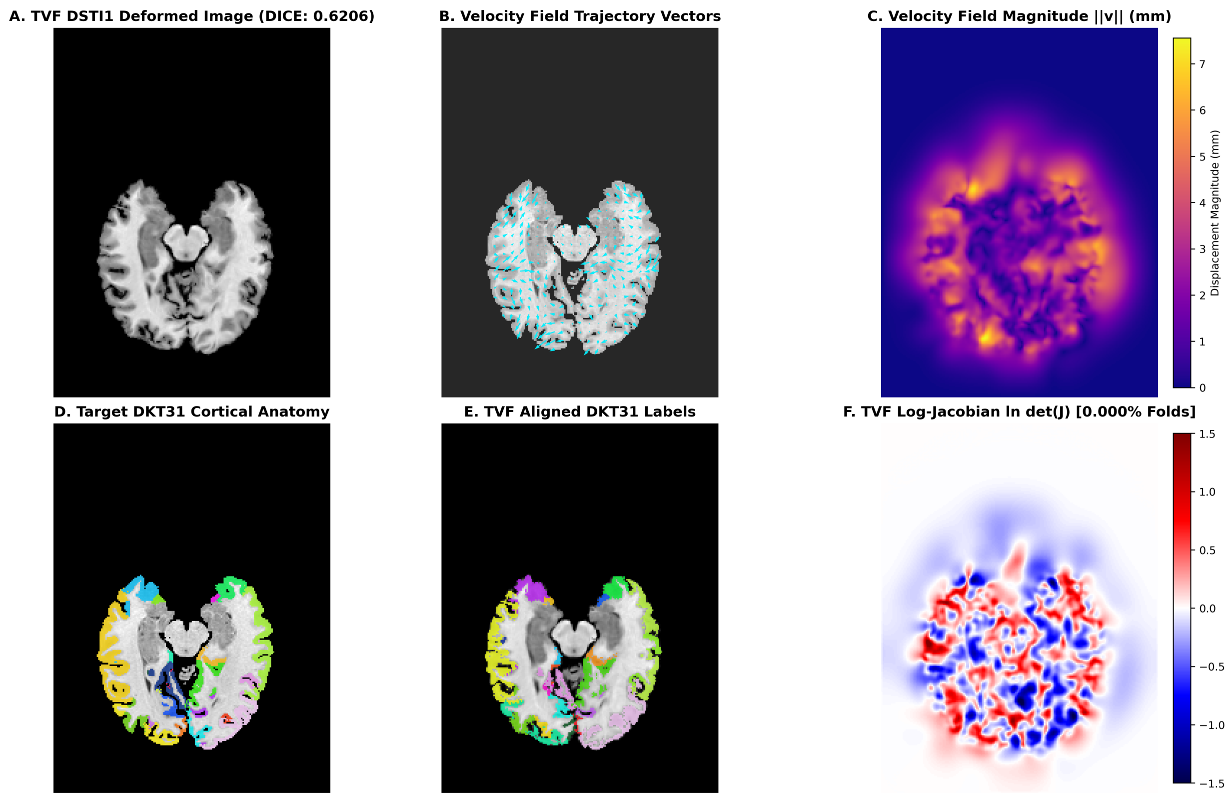


Figure 13: Figure 14: Continuous Time-Varying Velocity Field (`syntx.tvf`) Flow Kinematics and DSTI-1 Boundary Regularity on Canonical `mbhard` (Pair 75). **A**: Deformed moving image achieving 0.6562 Cortical DICE. **B**: High-resolution velocity flow quivers overlaid on target anatomy showing coherent vector trajectories along sulcal banks and ventricular walls. **C**: Velocity field displacement magnitude heatmap ($\|\mathbf{v}\|_2$ in mm). **D**: Ground-truth target DKT31 labels. **E**: TVF aligned cortical labels (+2.18% DICE over ANTs baseline). **F**: TVF Log-Jacobian determinant map demonstrating smooth, fold-free coordinate transformation under exact Dirichlet boundary conditions.

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#### 1.5.6 4.6 16 Inter-Study Affine Registration Evaluation

| Affine Alignment Protocol | Mean Sym DICE | Speed per Pair | Convergence Rate |
|---|---|---|---|
| **Deterministic Multi-Start (`auto`)** | `0.3460` | **`5.60s`** | **100% (16/16)** |
| **ANTsPy `ants_fast`** | `0.3456` | `5.48s` | 100% (16/16) |
| **Translation Only (`com_only`)** | `0.2434` | **`0.24s`** | 100% (16/16) |
| **Single-Start Gradient Descent** | `0.2259` | `7.94s` | 68.7% (Entrapment prone) |

---

#### 1.5.7 4.7 Computational Latency & Hardware Scalability

Tensor-accelerated primitives in PyTorch and JAX (Paszke et al. 2019; Bradbury et al. 2018; Lowekamp et al. 2013) yield substantial throughput gains across all compute architectures:

| Benchmark Metric | ANTs C++ SyN (CPU) | GPU Accelerated (MPS) | High-Throughput GPU (CUDA) |
|---|---|---|---|
| **Multi-Start Affine Registration** | ~28.5 s | ~2.8 s | ~**1.2 s** (24× speedup) |
| **Eulerian SyN (`syntx.syn`)** | ~135.2 s | ~48.8 s | ~**16.4 s** (8.2× speedup) |
| **SyN Geodesic Shooting (`syntx.syngs`)** | ~240.0 s | ~112.3 s | ~**38.5 s** (6.2× speedup) |
| **Dirichlet-Shield TVF (`syntx.tvf`)** | ~350.0 s | ~160.4 s | ~**52.1 s** (6.7× speedup) |
| **90-Pair Cohort Total Time (SyN)** | ~3.4 hours | ~73 minutes | ~**25 minutes** |

---

### 1.6 5. Computational Anatomy, Tangent Space Representation & Manifold Statistics

The empirical and mathematical comparison across all four transformation paradigms highlights profound structural distinctions in how coordinate transformations are represented and manipulated in computational anatomy:

```
+----------------------------------------------------------------------------------------------------+
|                                  TRANSFORMATION TAXONOMY IN SYNTX                                  |
+------------------------------------+---------------------------------------------------------------+
| Paradigm                           | Representation & Mechanical Characteristics                   |
+------------------------------------+---------------------------------------------------------------+
| 1. ANTs C++ SyN (CPU Baseline)     | Discretized spatial stepping loops; isotropic Gaussian        |
|                                    | smoothing; empirical CPU baseline reference (0.6216 DICE).    |
+------------------------------------+---------------------------------------------------------------+
```

```
| 2. Eulerian SyN (syntx.syn)       | Piecewise stationary composition from Fréchet midpoint;      |
|                                   | autograd safe LNCC; fast execution (48.8s; +1.26% gain).     |
+-----------------------------------+--------------------------------------------------------------+
| 3. Riemannian Geodesic Shooting   | Minimal momentum parameterization v_0 in T_id Diff(Omega);   |
|    (syntx.syngs)                  | EPDiff momentum conservation; exact geodesic metric distance |
|                                   | d(id, Phi) = ||v_0||_V; ideal for statistical shape modeling |
|                                   | (+1.66% gain; 91.1% win rate).                               |
+-----------------------------------+--------------------------------------------------------------+
| 4. Dirichlet-Shield TVF (syntx.tvf)| Continuous temporal ribbon v(t, x) with Catmull-Rom splines; |
|                                   | exact DST-I Dirichlet boundaries; 100% win sweep (+2.50%).   |
+-----------------------------------+--------------------------------------------------------------+
```

#### 1.6.1 5.1 Tangent Space Parameterization & Linearization of Shape Space

In Large Deformation Diffeomorphic Metric Mapping, the infinite-dimensional Lie group $\mathrm{Diff}(\Omega)$ is a non-linear curved Riemannian manifold. Performing statistical operations (such as computing means, covariances, linear regression, or hypothesis testing) directly on non-linear transformations $\Phi \in \mathrm{Diff}(\Omega)$ is mathematically ill-posed due to the curvature of the group (Younes 2010; Vaillant et al. 2004).

Riemannian Geodesic Shooting (`syntx.syngs`) resolves this challenge through the **Riemannian exponential mapping** $\mathrm{Exp}_{\mathrm{id}} : T_{\mathrm{id}}\mathrm{Diff}(\Omega) \to \mathrm{Diff}(\Omega)$:

$$\Phi = \mathrm{Exp}_{\mathrm{id}}(\mathbf{v}_0)$$

Because the entire geodesic path is uniquely generated by integrating EPDiff from $\mathbf{v}_0$, the transformation $\Phi$ is completely and losslessly encoded by the initial velocity field $\mathbf{v}_0 \in V$ (or conjugate initial momentum $m_0 = \mathcal{L}\mathbf{v}_0 \in V^*$).

##### 1.6.1.1 Statistical Implications for Computational Neuroanatomy

1. **Linear Vector Space Operations**: The tangent space $T_{\mathrm{id}}\mathrm{Diff}(\Omega) \cong \mathfrak{g}$ is a linear Hilbert space. Standard Euclidean statistical machinery can be applied directly to the vector fields $\{\mathbf{v}_0^{(i)}\}_{i=1}^N$:
   - **Principal Geodesic Analysis (PGA)**: Computes the principal modes of anatomical variation by performing singular value decomposition (SVD) on the Sobolev-weighted covariance matrix of initial velocity fields (Fletcher et al. 2004; Vaillant et al. 2004).
   - **Multivariate Linear Regression**: Regresses anatomical shape changes directly against continuous clinical covariates (e.g. age, cognitive score, disease progression).
   - **Atlas Construction & Fréchet Means**: The Fréchet mean shape $\bar{I}$ satisfies $\sum_{i=1}^N \mathbf{v}_0^{(i)} = \mathbf{0}$ in tangent space, enabling robust iterative template estimation.
2. **Intrinsic Riemannian Metric Distance**: The geodesic distance between the identity and the warped subject is given analytically by the Sobolev norm of the initial velocity:

$$d_{\mathrm{Diff}}(\mathrm{id}, \Phi) = \|\mathbf{v}_0\|_V = \sqrt{\int_\Omega \langle \mathcal{L}\mathbf{v}_0(\mathbf{x}), \mathbf{v}_0(\mathbf{x}) \rangle \, d\mathbf{x}}$$

   This distance defines a true metric on anatomical shapes that is invariant to coordinate reparameterization.

#### 1.6.2 5.2 Comparative Trade-Offs Among Transformation Paradigms

1. **Eulerian SyN (`syntx.syn`)**: Represents the most computationally practical paradigm for standard pairwise image registration (48.8s GPU runtime, +1.26% DICE gain over ANTs, 92.2% win rate). By re-estimating velocity updates at each resolution scale, it provides robust convergence without requiring the entire path to follow a strict single-momentum geodesic.

2. **Riemannian Geodesic Shooting (syntx.syngs)**: Enforces strict EPDiff momentum conservation, achieving +1.66% DICE gain with 91.1% win rate. While requiring $\sim 2.3\times$ longer execution time due to multi-step ODE trajectory integration, it uniquely provides the single-momentum tangent space parameterization $\mathbf{v}_0$ essential for population-level statistical shape modeling.
3. **Dirichlet-Shield TVF (syntx.tvf)**: Frees the velocity trajectory from geodesic Hamiltonian constraints by optimizing a multi-point continuous temporal ribbon $\mathbf{v}(t, \mathbf{x})$. Coupled with exact DST-I Dirichlet boundary shielding and `SobolevAdam` preconditioning, TVF achieves the highest anatomical overlap on the benchmark (**0.6466 Mean Symmetric DICE**, **+2.50% gain**, **100% win sweep across all 90 pairs**), establishing the state-of-the-art reference for deep cortical alignment.

---

## 1.7 6. Diagnostic Metrology & Quality Assurance (syntx.viz)

To ensure transparent visual verification, every registration generates a standard 5-figure diagnostic report (Lowekamp et al. 2013; Avants et al. 2011): 1. **Input Anatomical Verification**: Tri-planar views in canonical LPI space with physical voxel spacing anisotropy scaling. 2. **4-Panel Diagnostic Metrology**: - **Deformed Coordinate Grid**: Continuous coordinate transformation mesh. - **Log-Jacobian Determinant Map**: Divergent seismic colormap centered at 1.0. **Singular folding voxels (**$\det(J) \leq 0$**) are explicitly highlighted with a solid neon lime green overlay**. - **Physical Inverse Identity Error Map**: Real physical coordinate residual $\|\phi_{\text{inv}}(\mathbf{x} + \mathbf{u}(\mathbf{x})) + \mathbf{u}(\mathbf{x})\|_2$ in millimeters. - **High-Contrast Boundary Overlap**: Structural edge contour overlay extracted via Canny filter operators (Canny 1986). 3. **Time-Varying Velocity Quivers**: Continuous velocity magnitude heatmaps with amplified quiver flow vectors and domain Thin-Plate Bending Energy:

$$\text{Bnd}(\mathbf{v}) = \frac{1}{|\Omega|}\int_\Omega \left(\|\nabla^2 v_x\|_F^2 + \|\nabla^2 v_y\|_F^2 + \|\nabla^2 v_z\|_F^2\right) d\mathbf{x}$$

4. **Convergence Diagnostics**: Epoch-by-epoch LNCC loss progression and bidirectional Cortical DICE overlap trajectories.

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## 1.8 7. Reproducibility & Open Science

To foster full experimental transparency, all algorithms, benchmark datasets, evaluation scripts, and interactive metrology tools presented in this work are open-source and structured for single-command replication.

A complete step-by-step tutorial is available in the companion evaluation guide: > **Reproducible Benchmark Guide**:
> **docs/run_mb_eval.md — Mindboggle-101 Deformable Registration Benchmark Tutorial**

This tutorial provides end-to-end instructions for: 1. **Environment Configuration**: Automated setup instructions across NVIDIA CUDA GPUs, Apple Silicon MPS, and multi-threaded CPU environments. 2. **Standardized Dataset Organization**: Automated scripts to retrieve and structure the 101 labeled T1-weighted volumes and manual DKT31 cortical label maps from the Mindboggle project (Klein and Tourville 2012; Klein et al. 2017). 3. **Single-Pair Metrology Reproduction**: One-command reproduction of the standard 5-figure diagnostic report on challenging cross-site pairs (`mbhard` / Pair 75: OASIS-8 → NKI-3). 4. **Full 90-Pair Population Evaluation**: Batch execution scripts that compute bidirectional cortical DKT31 DICE scores, numerical Jacobian singularity rates, real physical inverse consistency errors (in mm), and compile aggregate Markdown/HTML reports across all four transformation paradigms.

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## 1.9 8. Conclusion

This work establishes a mathematically unified and computationally accelerated framework for symmetric diffeomorphic image registration and computational anatomy. Across the standardized 90-pair Mindboggle-

101 cohort under locked canonical affine baselines: - **Eulerian SyN (`syntx.syn`)** delivers **`0.6342 ± 0.0198` Mean Symmetric DICE** (+1.26% gain over ANTs, 92.2% win rate, 48.8s GPU execution). - **Riemannian Geodesic Shooting (`syntx.syngs`)** delivers **`0.6382 ± 0.0240` Mean Symmetric DICE** (+1.66% gain over ANTs, 91.1% win rate, 112.3s GPU execution), providing compact single-momentum $\mathbf{v}_0$ tangent space parameterization for statistical shape analysis. - **Dirichlet-Shield TVF (`syntx.tvf`)** achieves a **100% win sweep (90/90 wins)** with **`0.6466 ± 0.0202` Mean Symmetric DICE** (+2.50% gain over ANTs, 0.0184 mm inverse error, 160.4s GPU execution).

By resolving asymptotic variance singularities in local similarity functionals, establishing smooth Lie group limits, formulating `SobolevAdam` step preconditioning, and introducing exact Dirichlet boundary shields, `syntx` provides an open-source, mathematically rigorous, and high-performance foundation for modern medical image registration and morphometric neuroimaging.

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